

Pharmacokinetics

Pharmacokinetics

- Pharmacokinetics is derived from two words: pharmaco – (drug) and kinesis – movement.

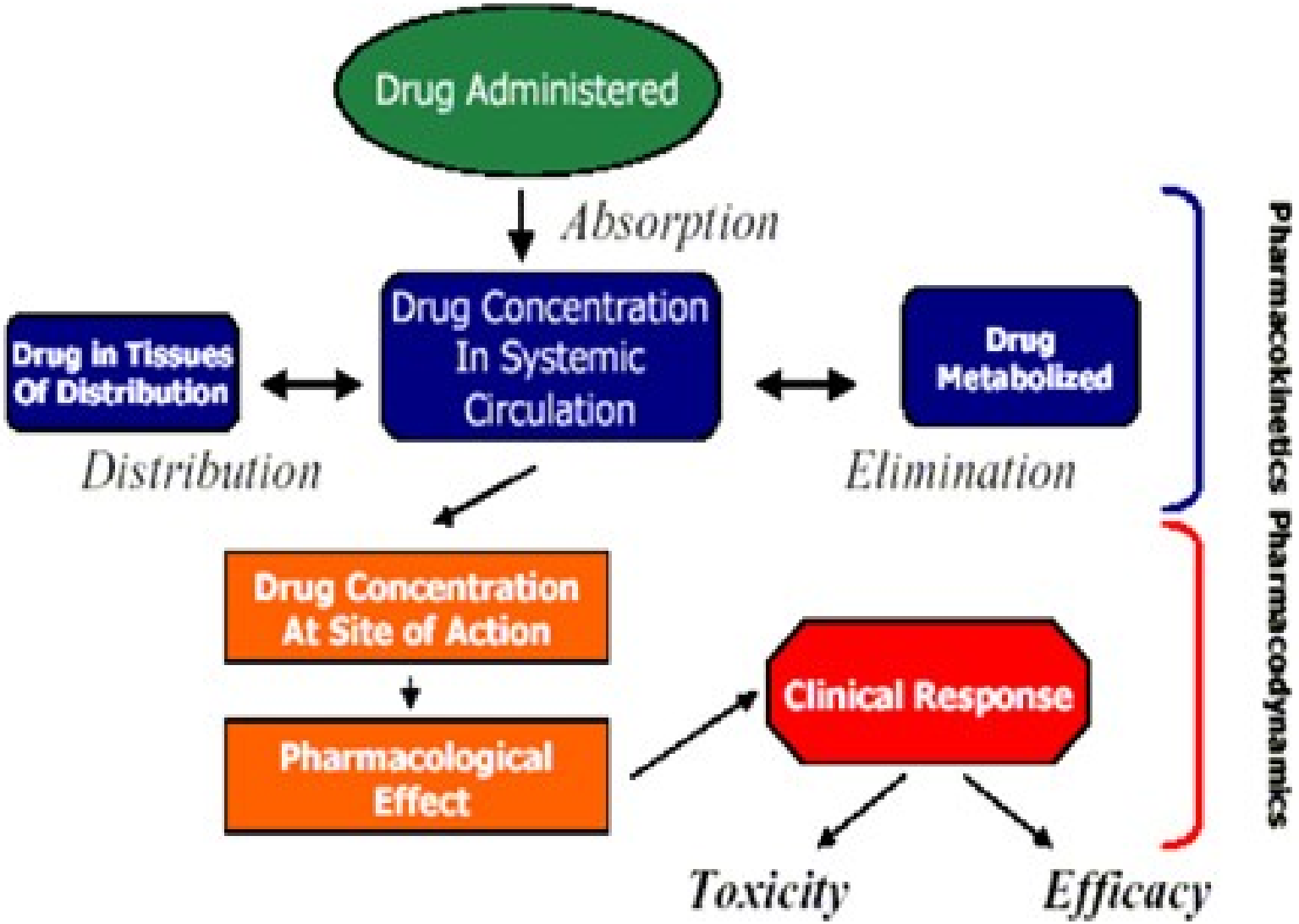
- In short, it is ` **what the body does to the drug`.**

- It includes absorption, distribution, metabolism, and excretion of a drug (ADME).

- **Why do we study pharmacokinetics?**
- The pharmacokinetic properties of a drug determine:
  1. **Onset of drug action**
  2. **Intensity of the drug's effect**
  3. **Duration of drug action**

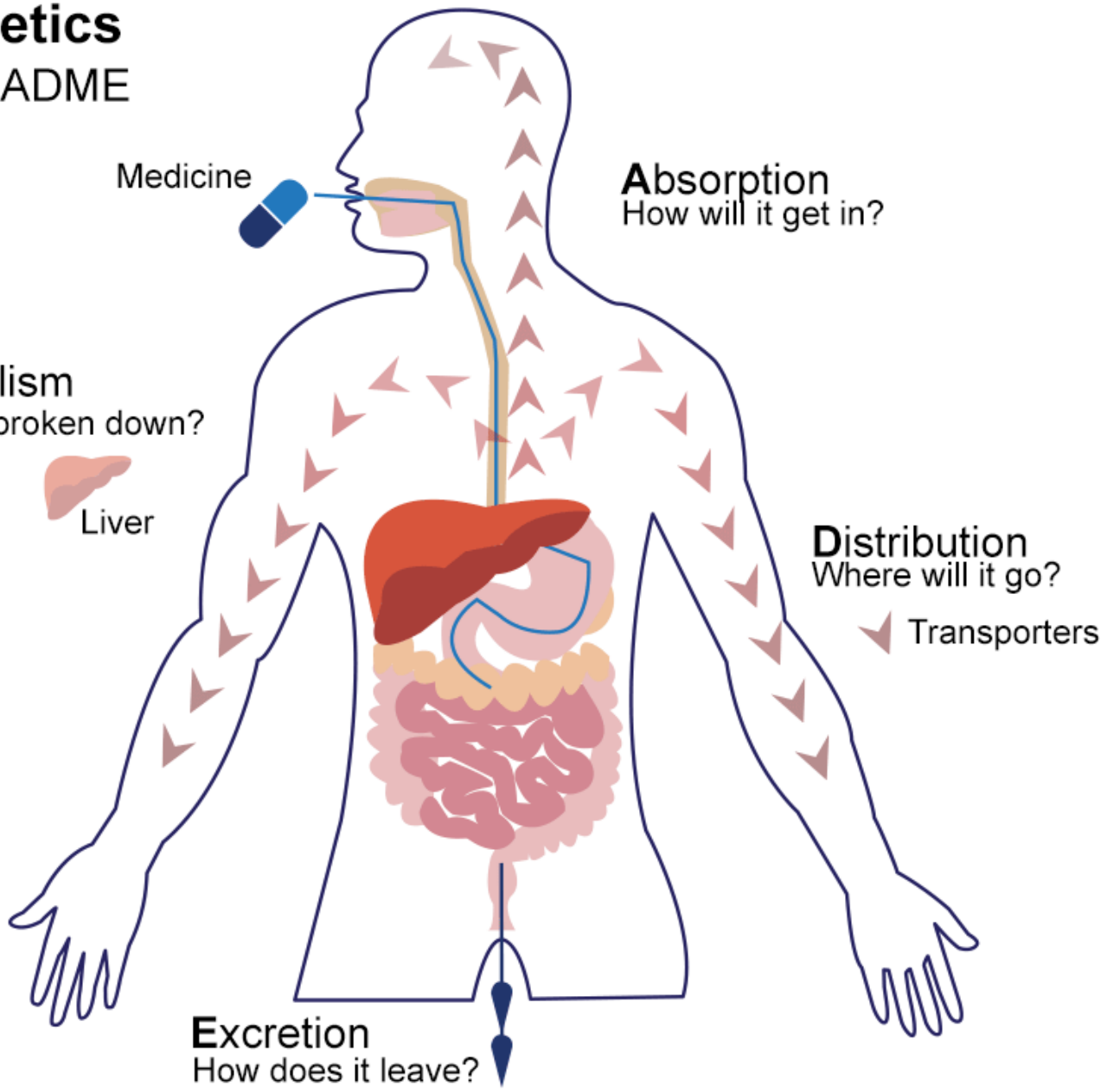
- This allow the clinician to design and optimize treatment regimens, including decisions as to:
  1. **Route of administration**
  2. **Dose and frequency of each dose**
  3. **Duration of treatment**

Pharmacodynamic vs. Pharmacokinetic



Pharmacokinetics

The principles of ADME



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I. ABSORPTION OF DRUGS

Absorption: is the transfer of a drug from its site of administration to the bloodstream via one of several mechanisms. (**except for drugs that are applied directly to the target tissue)**

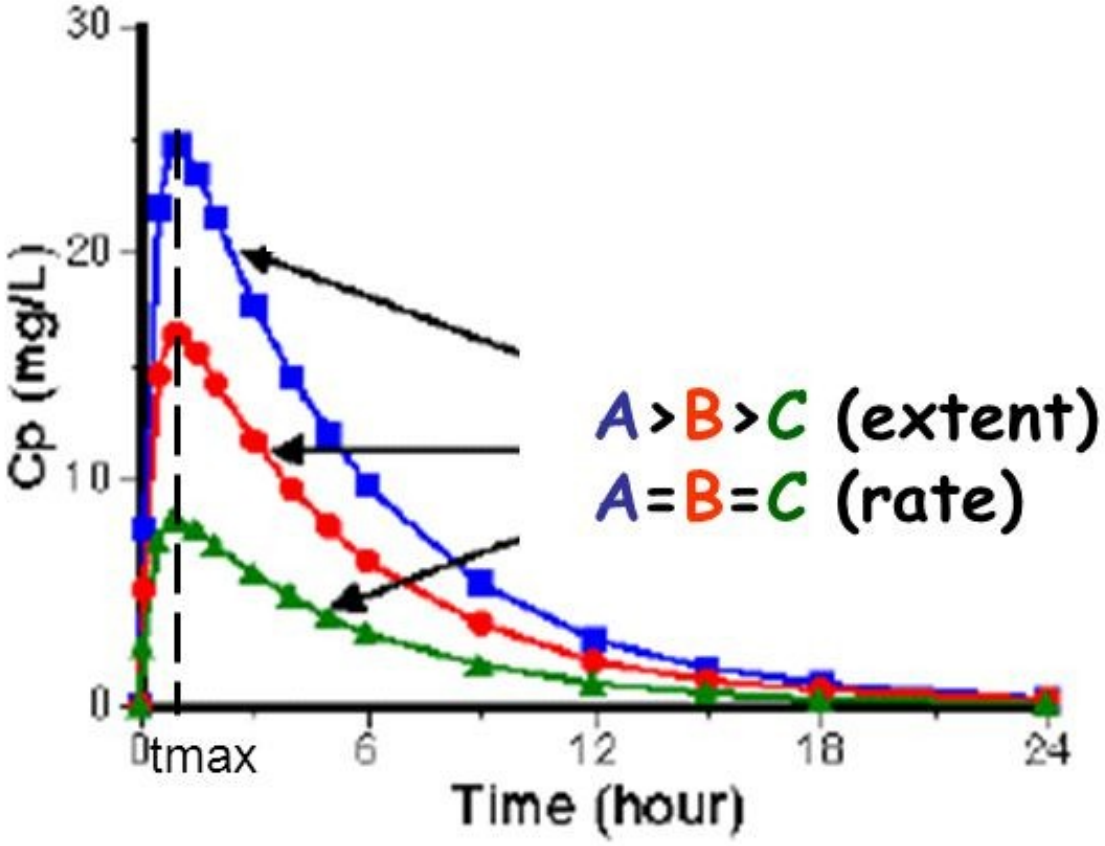
- **The rate and extent of absorption depend on:**
  1. Site of absorption
     - ✓ Route of administration
  2. The **chemical characteristics** of the drug

Absorption: Rate & Extent
(Bioavailability = %F)

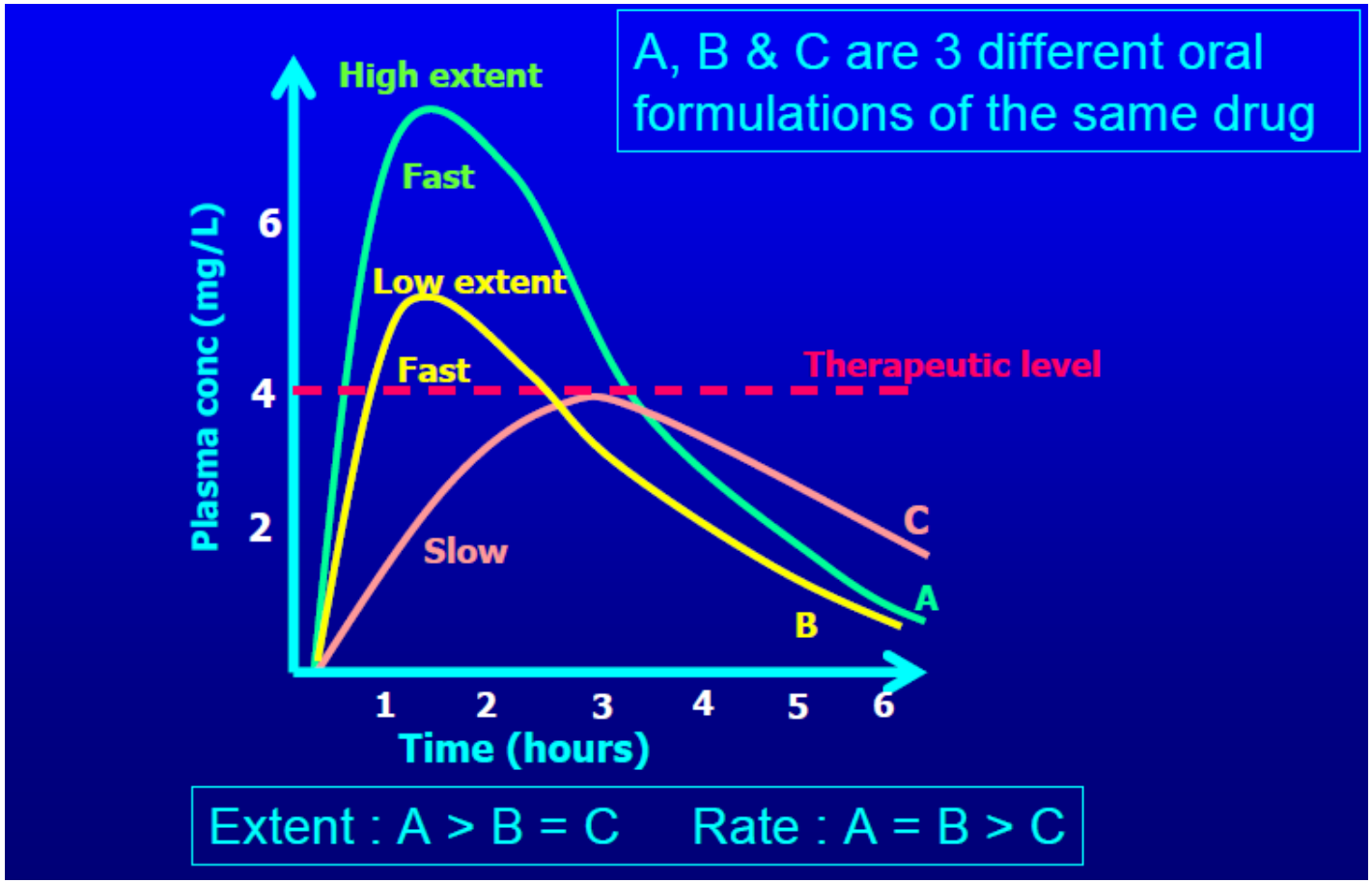
- **Rate:** how rapidly does the drug get from its site of administration to the general circulation ?

- **Extent:** How much of the administered dose enters the general circulation ?

Rate versus extent of absorption



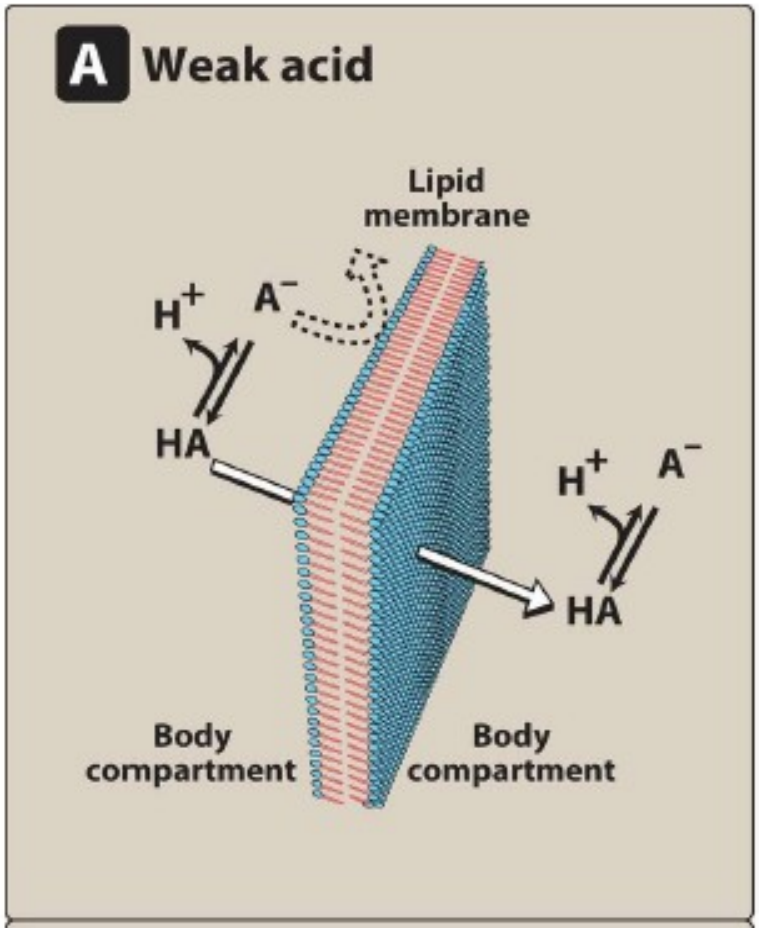
Rate versus extent of absorption

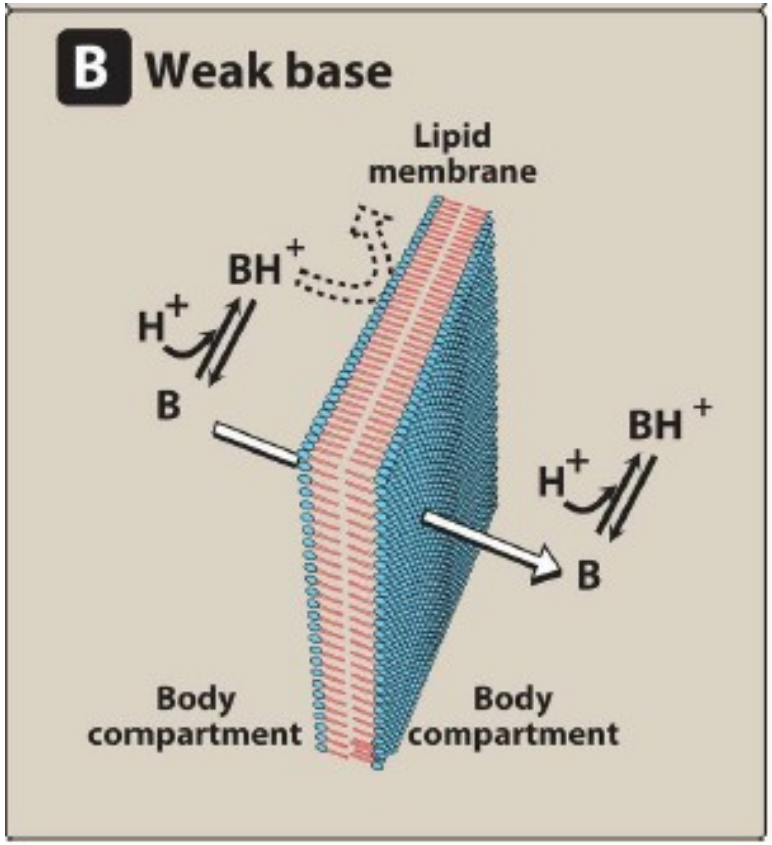


Factors influencing absorption

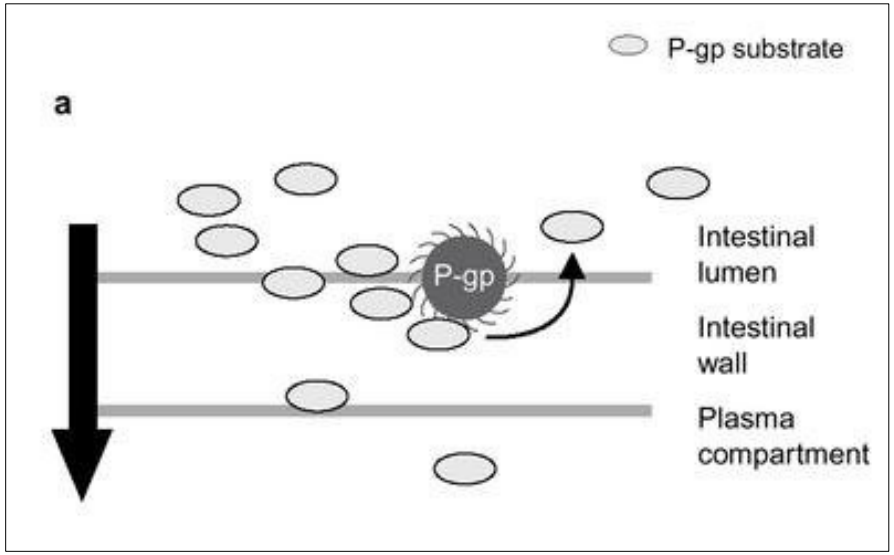
1. Effect of pH on drug absorption
2. Blood flow to the absorption site
3. Total surface area available for absorption
4. Contact time at the absorption surface
5. Expression of P-glycoprotein

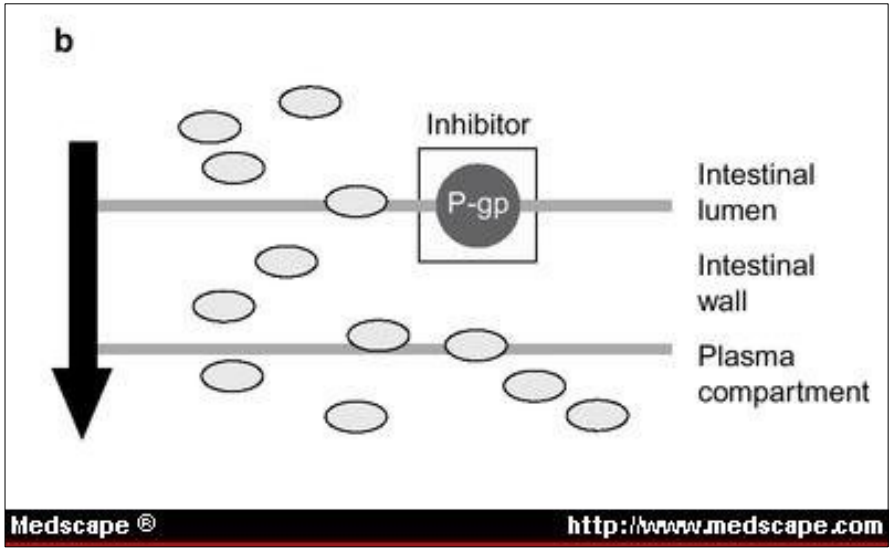
Effect of pH on drug absorption



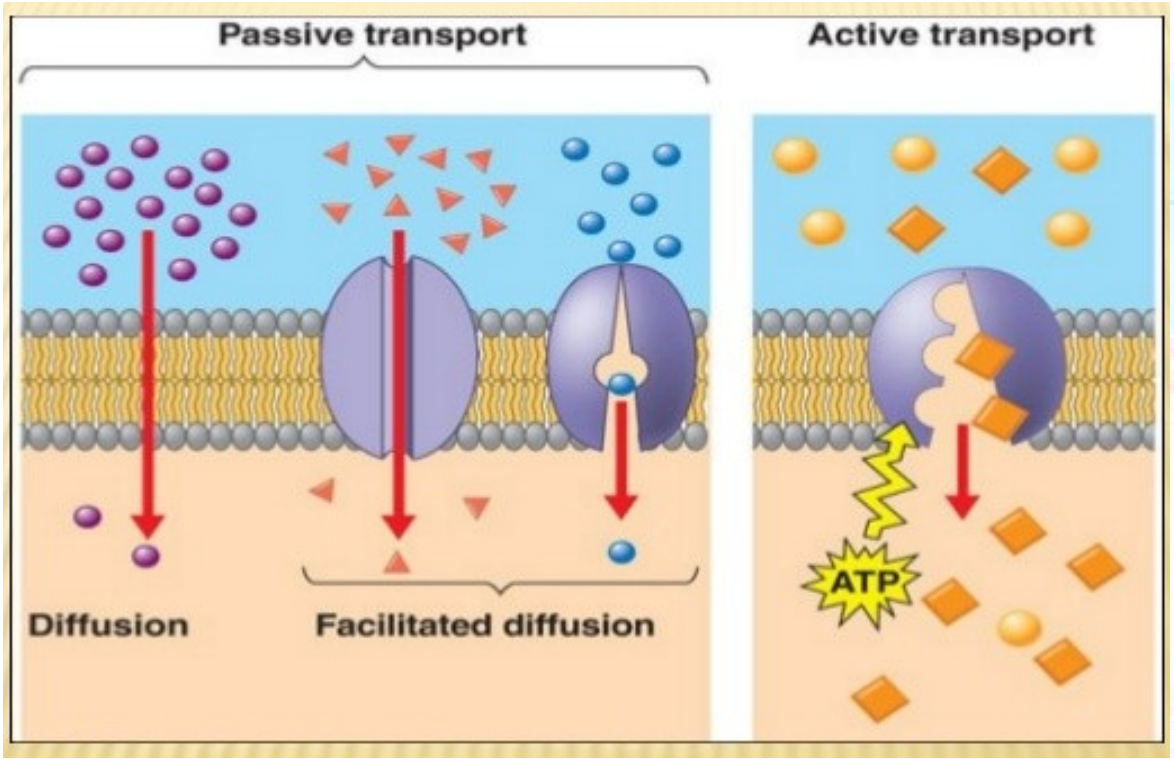


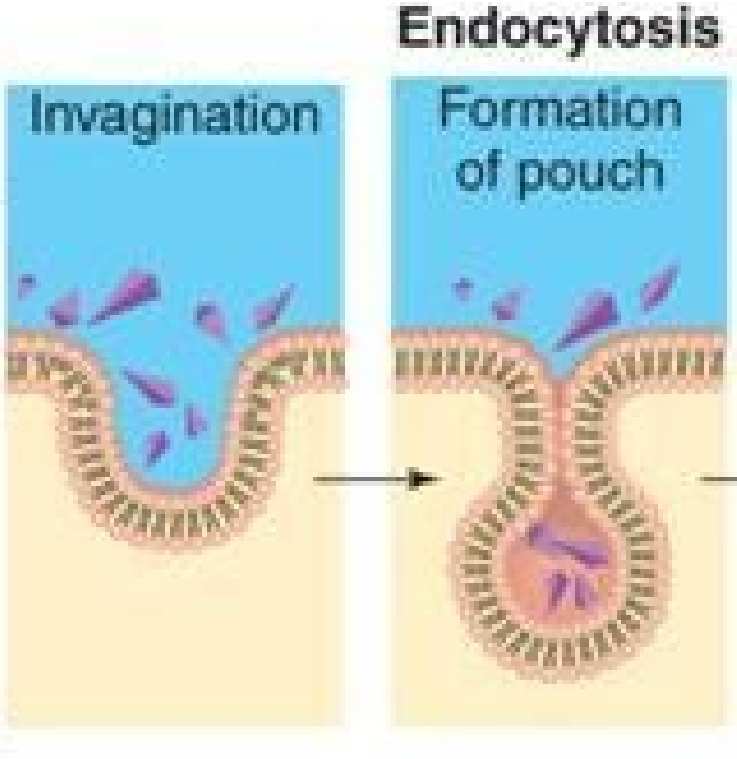
Expression of P-glycoprotein





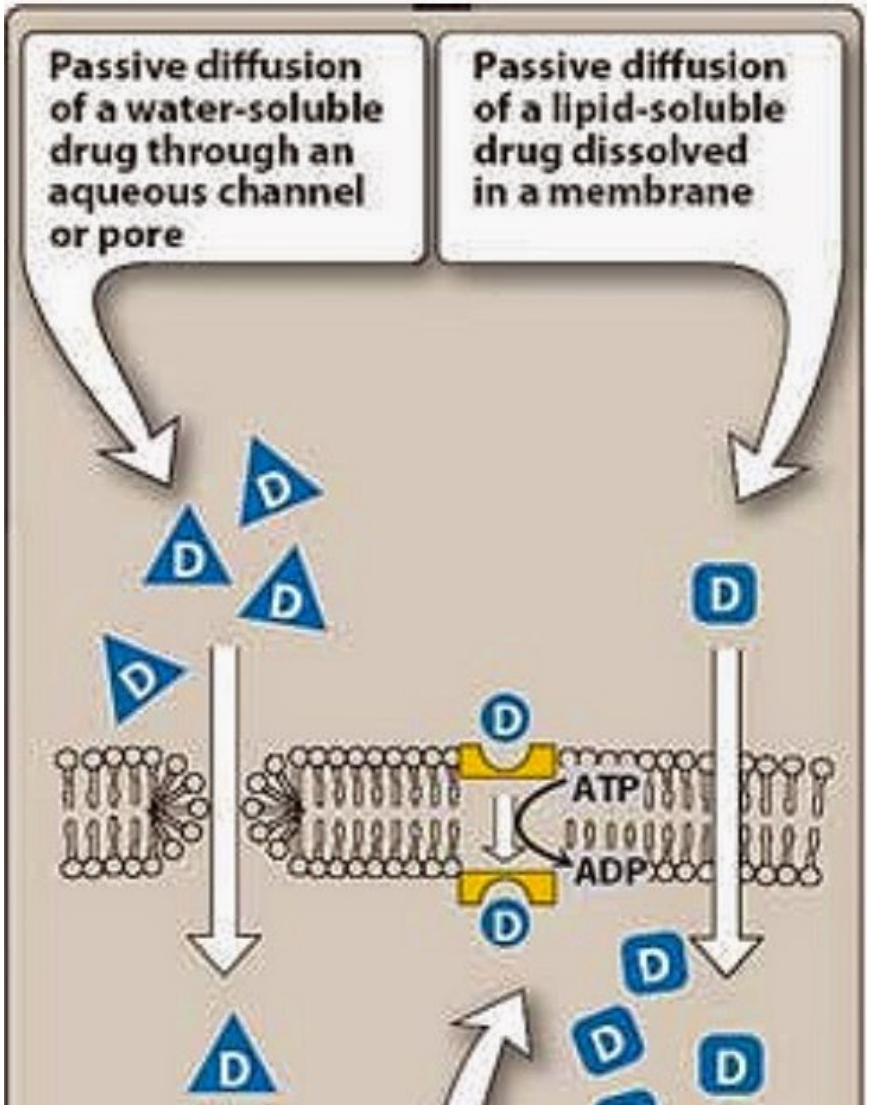
Mechanisms of absorption of drugs from the GI tract





1. **passive diffusion:** Movement of a solute through a biological barrier from the phase of higher concentration to phase of lower concentration (No need of energy and carrier)

e.g. highly lipid soluble drugs



2. **Facilitated diffusion**: It means the passage of drug across the biological membrane along the **concentration gradient** by the **protein carrier** mediated system also called as carrier mediated diffusion
3. **Active transport:** The process by which drugs pass across the biological membrane most often **against their concentration gradient** with the help of carriers along with the expenditure of **energy**
4. **Endocytosis:** It is the process by which the **large molecules** are engulfed by the cell membrane and releases them intracellularly

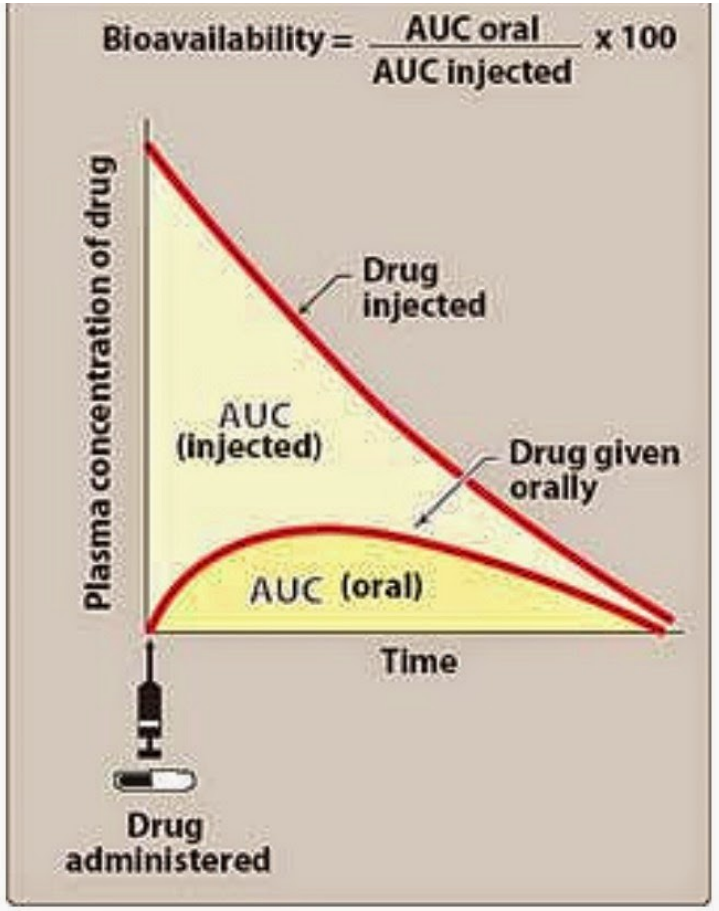
Bioavailability

- **Bioavailability is the fraction of administered drug that reaches the systemic circulation.**
- **Or, it is the rate and extent to which an administrated drug reaches the systemic circulation**

- For example
  - if 100 mg of a drug are administered orally
  - 70 mg of this drug are absorbed unchanged
  - the bioavailability is 0.7, or 70 percent.

Determination of bioavailability

- Bioavailability is determined by comparing **plasma levels of a drug** after a particular route of administration (for example, oral administration) with **plasma drug levels achieved by IV injection**
- Determining bioavailability is important for **calculating drug dosages** for non-intravenous routes of administration



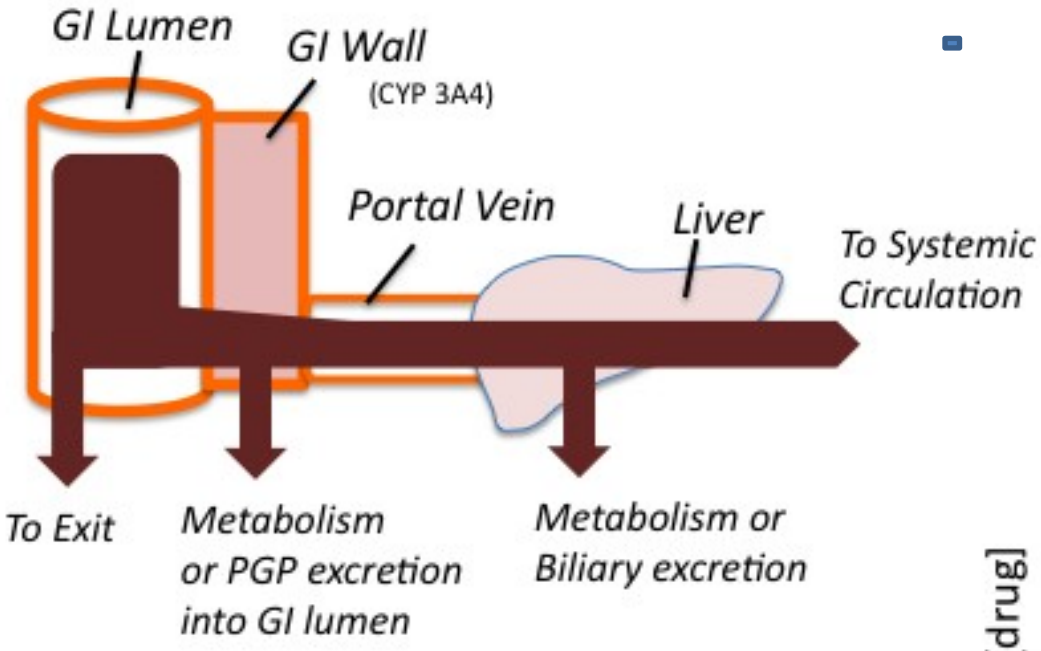
Factors that influence bioavailability

1- First-pass hepatic metabolism

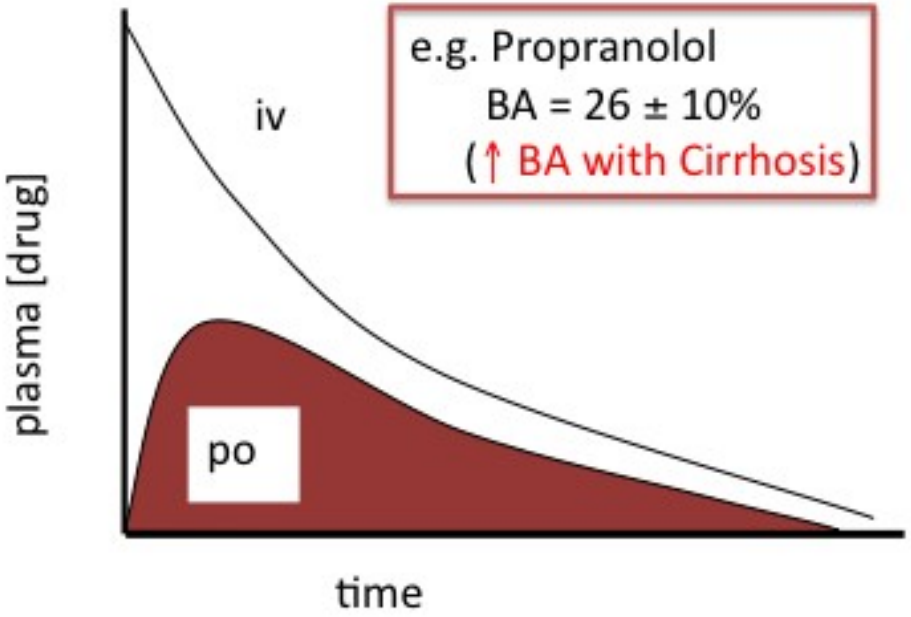
If the drug is rapidly metabolized by the liver, the amount of unchanged drug that gains access to the systemic circulation is decreased

2- Solubility of the drug:

- **Very hydrophilic drugs are poorly absorbed** because of their inability to cross the lipid-rich cell membranes
- **Extremely lipophilic drugs are also poorly absorbed**, because they are totally insoluble in aqueous body fluids and, and cannot gain access to the surface of cells
- drug to be readily absorbed, it must be **largely lipophilic**, and have some **solubility in aqueous**



$$BA = \frac{AUC_{po}}{AUC_{iv}} \times 100$$



Factors that influence bioavailability

3. Chemical instability

- Some drugs, such as penicillin G, are unstable in the pH of the gastric contents
- Others, such as insulin, are destroyed in the GI tract by degradative enzymes

4. Nature of the drug formulation

Drug absorption may be altered by factors unrelated to the chemistry of the drug

For example, particle size, salt form, enteric coatings and the presence of excipients can influence the ease of dissolution and, therefore, alter the rate of absorption.

Bioequivalence

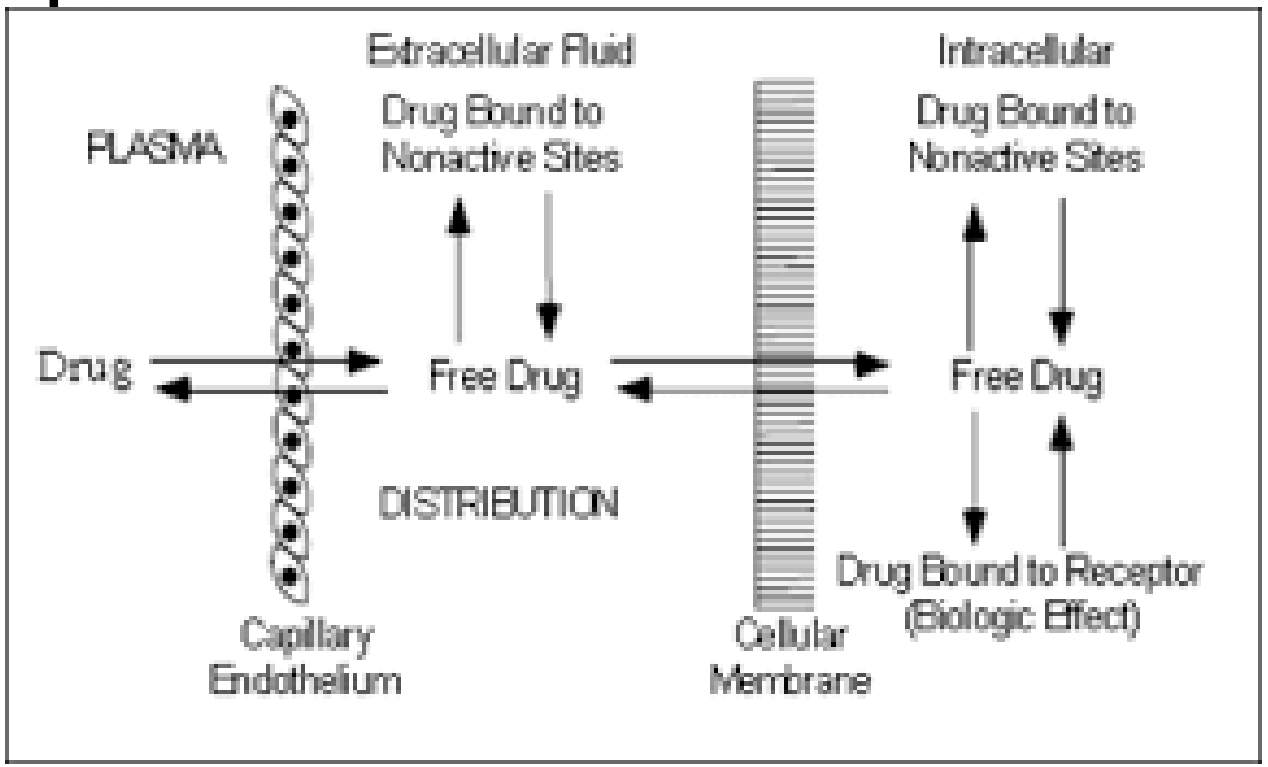
- Two related drug preparations are bioequivalent if:
  1. They show comparable bioavailability
  2. Similar times to achieve peak blood concentrations.

Therapeutic equivalence

- Two similar drug products are therapeutically equal if they are **pharmaceutically equivalent** with **similar clinical and safety profiles**

II. DRUG DISTRIBUTIOI

- **Distribution:** is defined as the reversible transfer of drugs between body fluid compartment



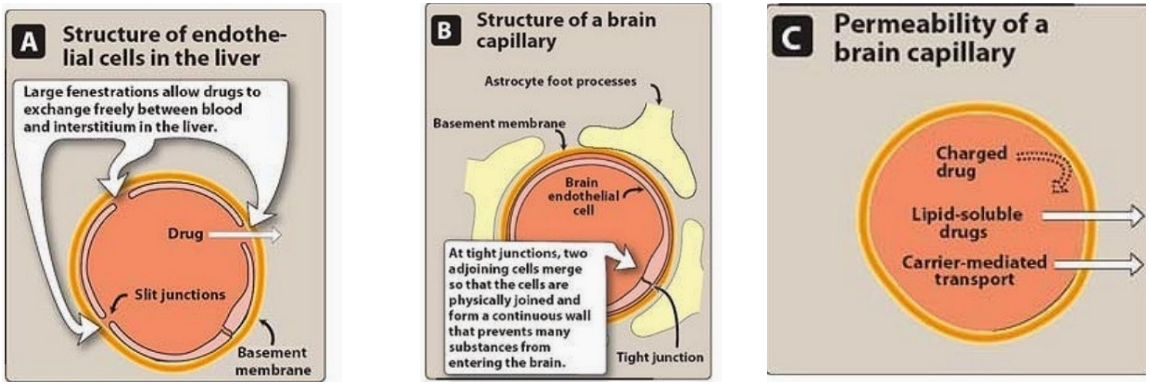
Factors affecting drug distribution

- The delivery of a drug from the plasma to the interstitial primarily depends on:

A. Cardiac output and regional blood flow
B. Capillary permeability
C. The tissue volume
D. The degree of binding of the drug to plasma and tissue proteins
E. lipophilicity of the drug

Factors affecting Capillary permeability

1. Capillary structure



2. Drug structure

- Hydrophobic drugs are readily move across most biologic membranes.
- hydrophilic drugs are not readily penetrate cell membranes, and must go through the slit junctions.

Binding of drugs to plasma proteins:

1. **Binding to plasma proteins:** Reversible binding to plasma proteins (as albumin) sequesters drugs in a nondiffusible form and slows their transfer out of the vascular compartment
2. **Binding to tissue proteins:** many drugs accumulate in tissues (binding to lipids, proteins, or nucleic acids), leading to higher concentration in tissues than in the extracellular fluid and blood
   - Prolong drug action
   - Local drug toxicity

Volume of distribution:

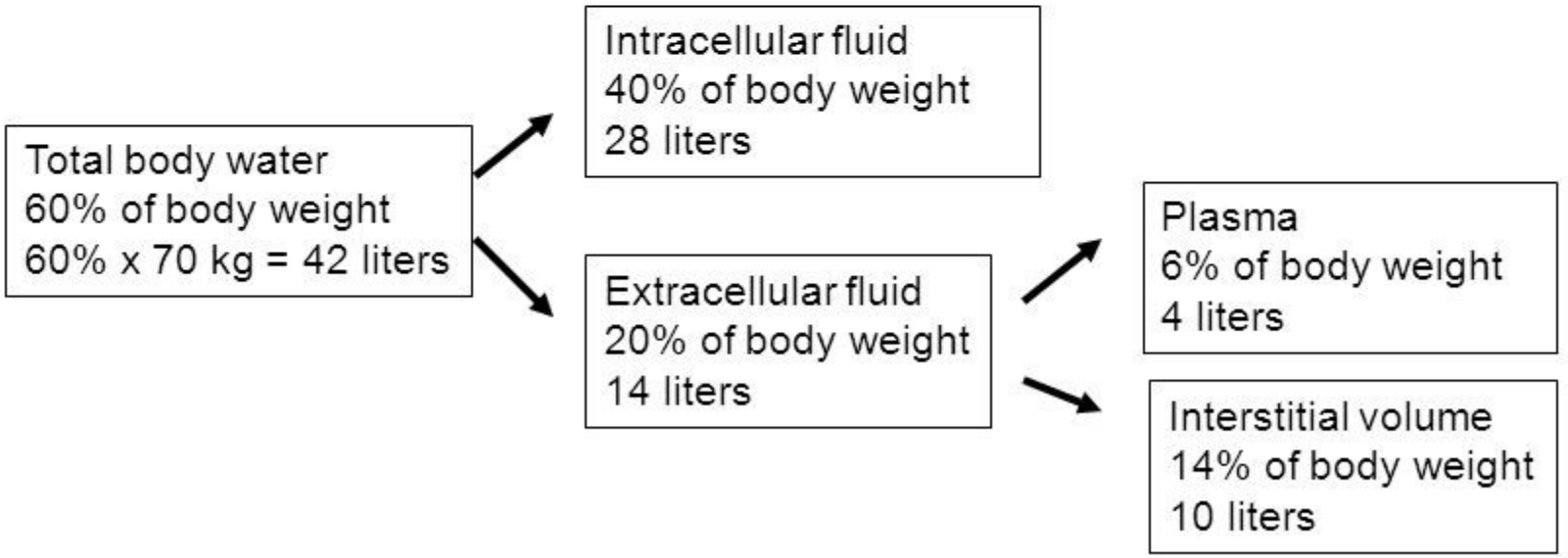
- The **volume of distribution** (V_d), = **apparent volume of distribution** is defined as
  - The distribution of a medication between plasma and the rest of the body

  Or,
  - **Apparent or hypothetical volume of fluid into which a drug is homogeneously distributed in the body**

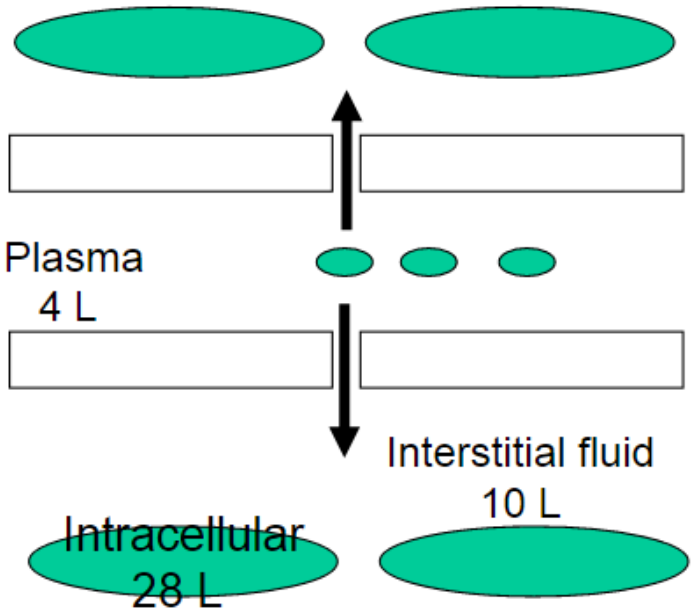
- **V_d** is a property of the drug and is determined by the drug's manufacturer

Volume of distribution (Vd): A hypothetical volume of fluid into which the drug is distributed.

- Water compartments in the body - three functionally distinct water compartments – plasma, interstitial fluid and intracellular fluid.



Drug Distribution



Plasma compartment: Large molecular weight or binds extensively to plasma proteins - trapped in the plasma compartment – Vd = 4 liters.

Extracellular fluid: Low molecular weight, hydrophilic - able to move through fenestrae into the interstitial fluid, but cannot move across the cell membranes into the intracellular fluid - sum of the plasma water and interstitial fluid – Vd = 14 liters.

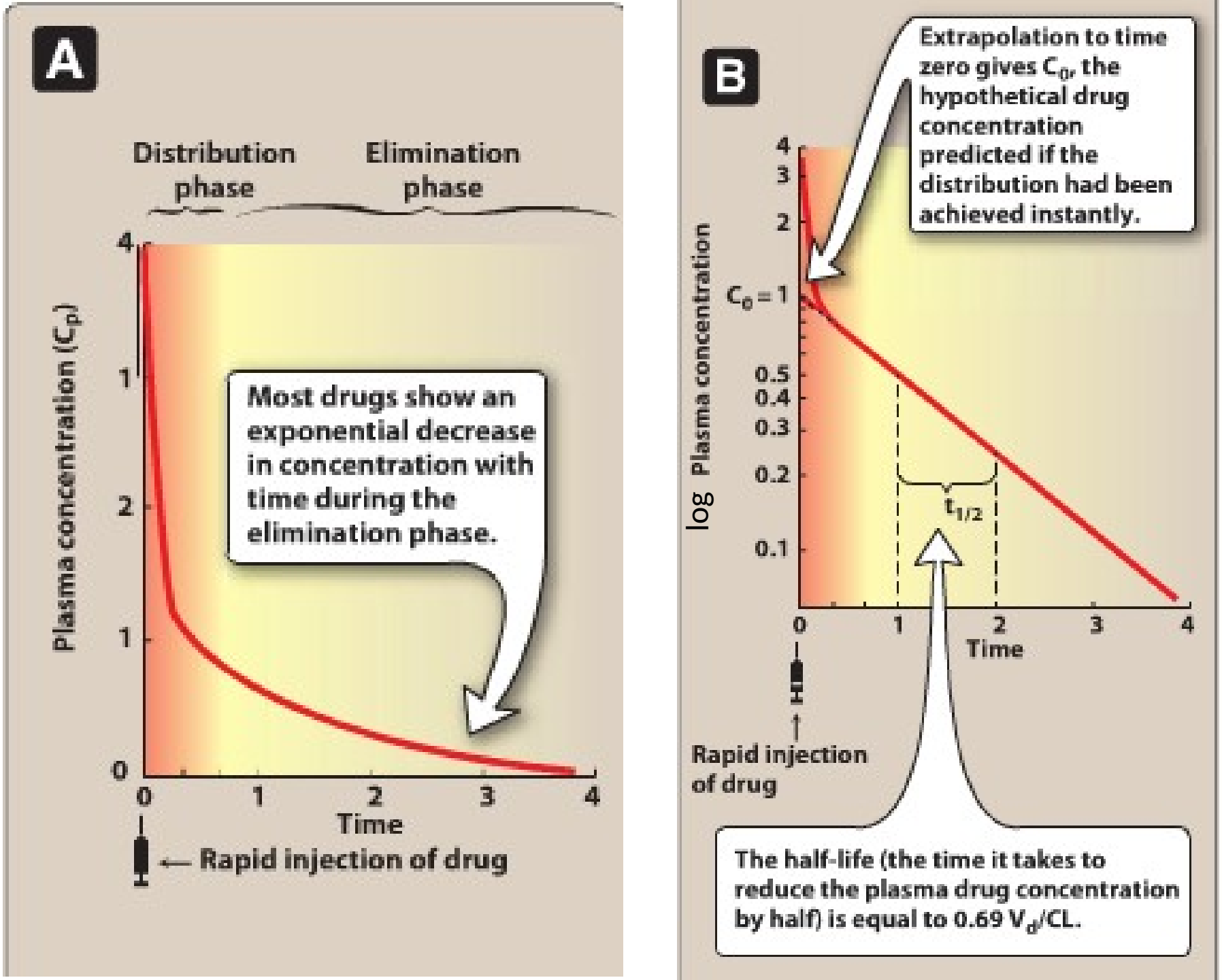
Total body water: Low molecular weight, hydrophobic - move into the interstitial fluid and the intracellular fluid – Vd = 42 liters.

Determination of V_d

$$V_D = \frac{\text{total amount of drug in the body}}{\text{drug blood plasma concentration}} \qquad V_d = \frac{\text{Amount of drug in the body}}{C_0}$$

- V_d is useful to compare the distribution of a drug with the volumes of the water compartments in the body.
- V_d is useful for calculating loading dose of a drug

- If V_d is big ⟹ the drug is more diluted than it should be (in the blood plasma), meaning more of it is distributed in tissue (i.e. not in plasma).


A
Distribution phase
Elimination phase
Plasma concentration (C_p)
4
1
2
1
0
Most drugs show an exponential decrease in concentration with time during the elimination phase.
0
1
2
3
4
Time
Rapid injection of drug
B
Extrapolation to time zero gives C_0, the hypothetical drug concentration predicted if the distribution had been achieved instantly.
log Plasma concentration
4
3
2
$C_0 = 1$
0.5
0.4
0.3
0.2
0.1
$t_{1/2}$
0
1
2
3
4
Time
Rapid injection of drug
The half-life (the time it takes to reduce the plasma drug concentration by half) is equal to 0.69 V_d/CL.

Effect of V_d on drug half-life:

- V_d has an important influence on the half-life of a drug
  - As drug elimination depends on the amount of drug delivered to the organs of metabolism per unit of time
- Larg V_d for a drug, most of the drug is in the extraplasmic space and is unavailable to the excretory organs.
- Any factor that increases V_d can lead to:
  - An increase in the half-life
  - Extend the duration of action.
- Delivery of drug to the organs of elimination depends on:
  - Blood flow
  - The fraction of the drug in the plasma.

III. DRUG METABOLISM

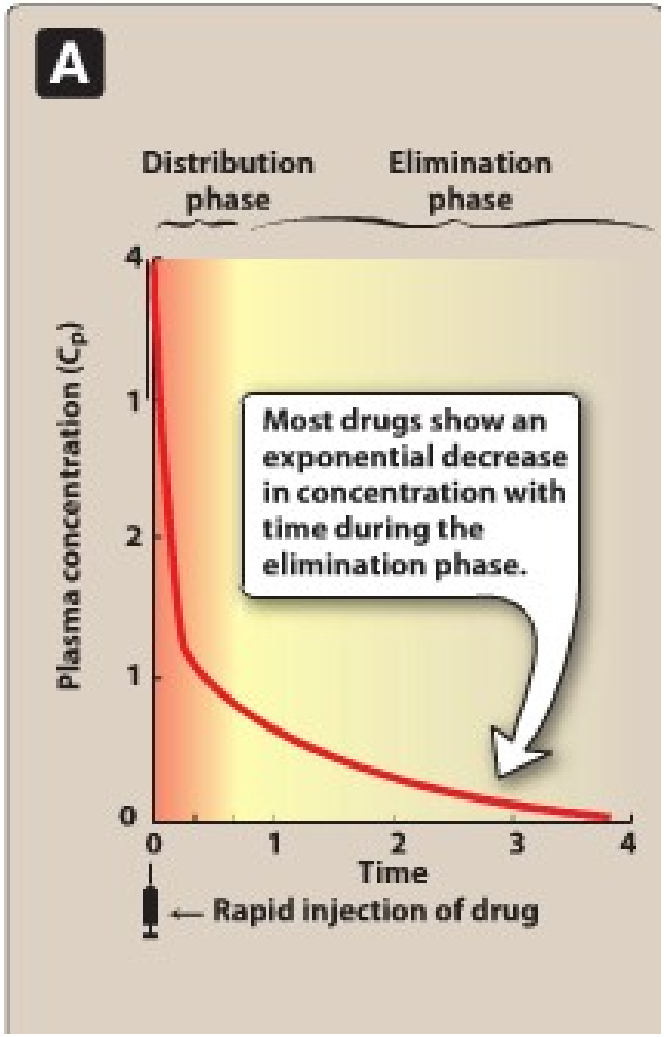
Elimination

- Drug elimination is usually divided into two major components:
  - *Drug* ***Biotransformation*** or *drug metabolism* is the process by which the drug is chemically converted in the body to a metabolite.
  - *Drug* ***Excretion*** is the removal of the intact drug.

- The process of elimination begins as soon as the drug enters the body.

- Three major routes of elimination:
  1) Hepatic metabolism
  2) Excretion into bile
  3) Excretion into the urine.

- Elimination causes the plasma concentration of a drug to decrease exponentially.
- A constant fraction of the drug is eliminated in a unit of time.

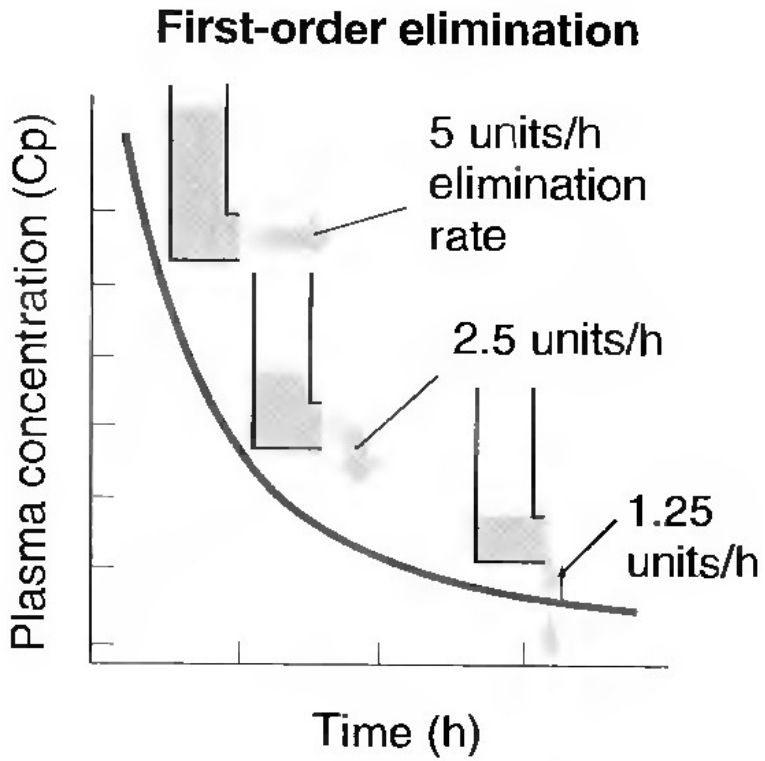


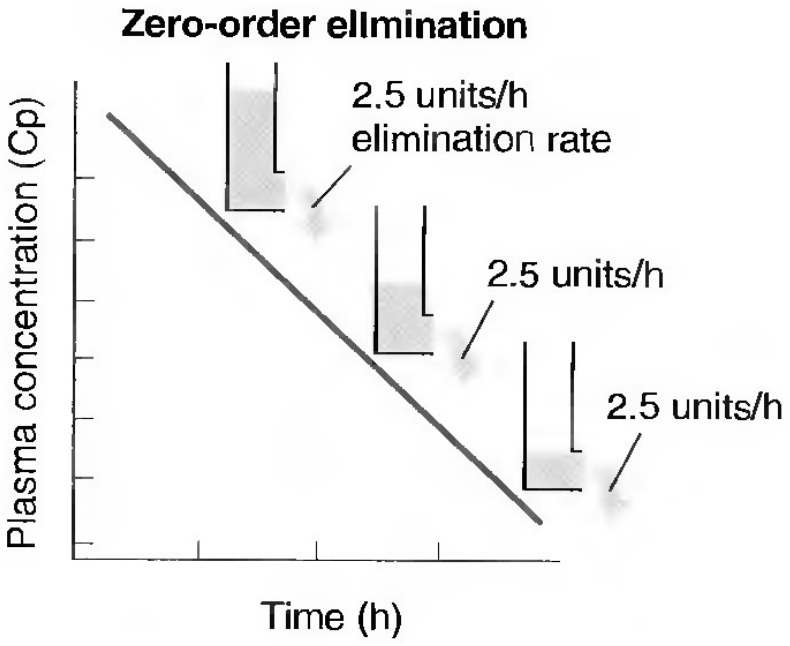
- **Metabolism** leads to products with increased polarity, which will allow the drug to be eliminated.

- **Clearance (CL)** estimates the amount of drug cleared from the body per unit of time

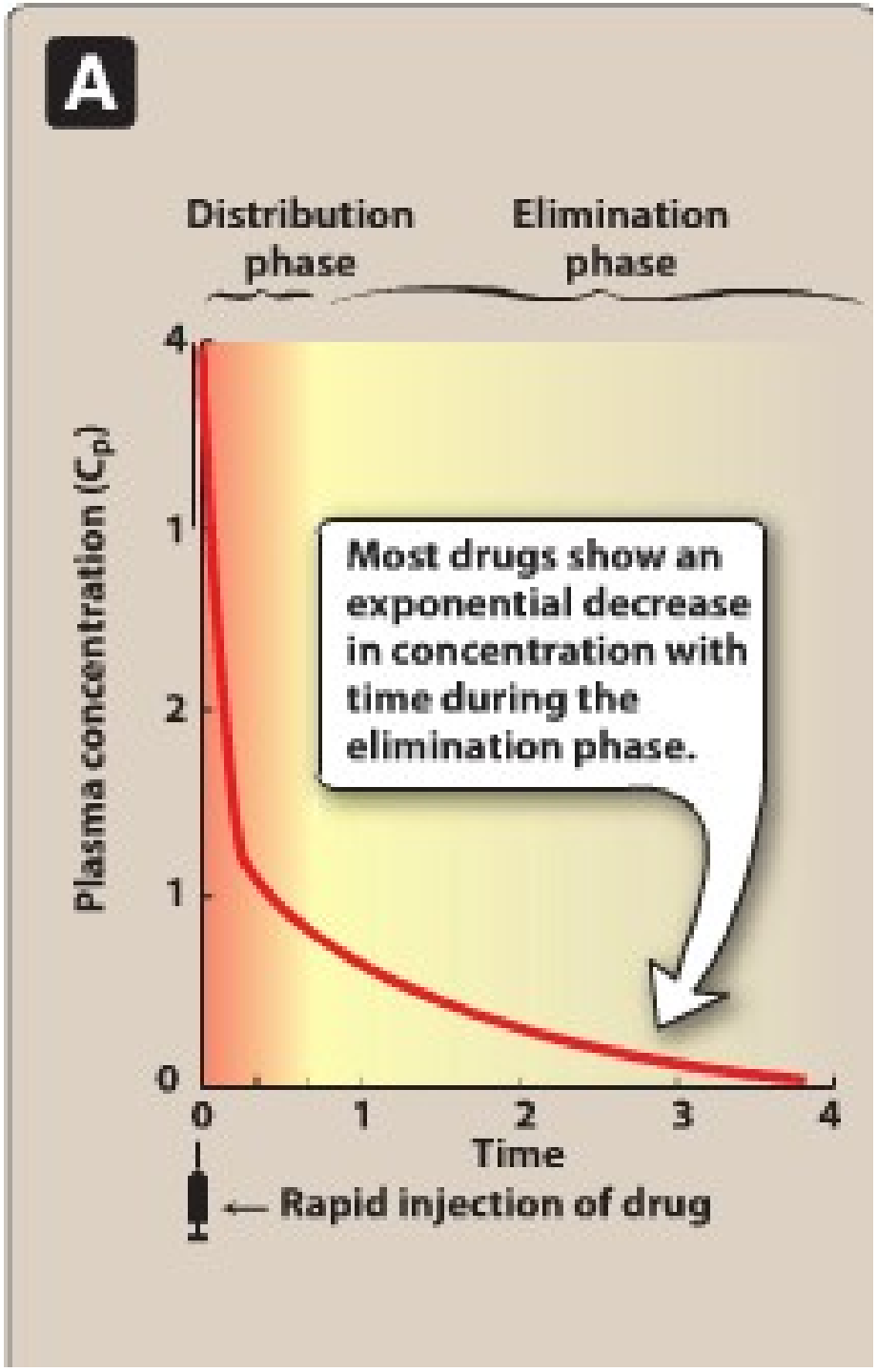
- Elimination could be through
  1. first-order kinetics: most drugs
  2. zero-order or non-linear kinetics: e.g. aspirin in high doses

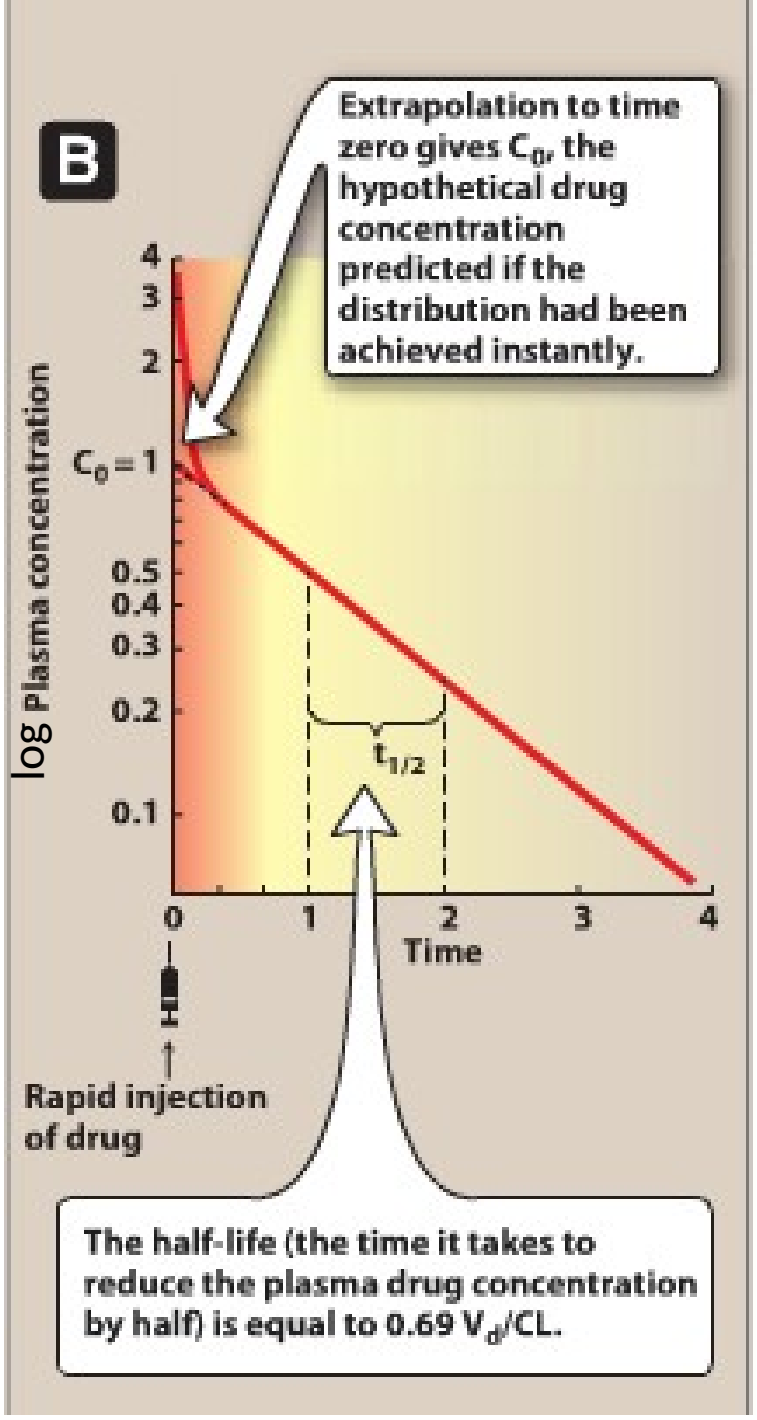
1st order elimination vs. zero order elimination





Drug half-life is often used as a measurement of drug CL, as , **V_d is a constant** for many drugs.


A
Distribution phase
Elimination phase
Plasma concentration (C_p)
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3
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0
Most drugs show an exponential decrease in concentration with time during the elimination phase.
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Time
Rapid injection of drug


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Extrapolation to time zero gives C_0, the hypothetical drug concentration predicted if the distribution had been achieved instantly.
log Plasma concentration
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3
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Time
Rapid injection of drug
The half-life (the time it takes to reduce the plasma drug concentration by half) is equal to 0.69 V_d/CL.

Drug metabolism

- The metabolism of a drug usually converts the lipid-soluble and un ionized compounds into water-soluble and ionized compound to be excreted.
- **Sites:**
  - Liver (main site)
  - Others: GI tract, kidney, blood, placenta
- Inactivation
- Activation
- Active drug to active metabolite

Reactions of drug metabolism

- Two general sets of reactions
  - **Phase I**
  - **Phase II**

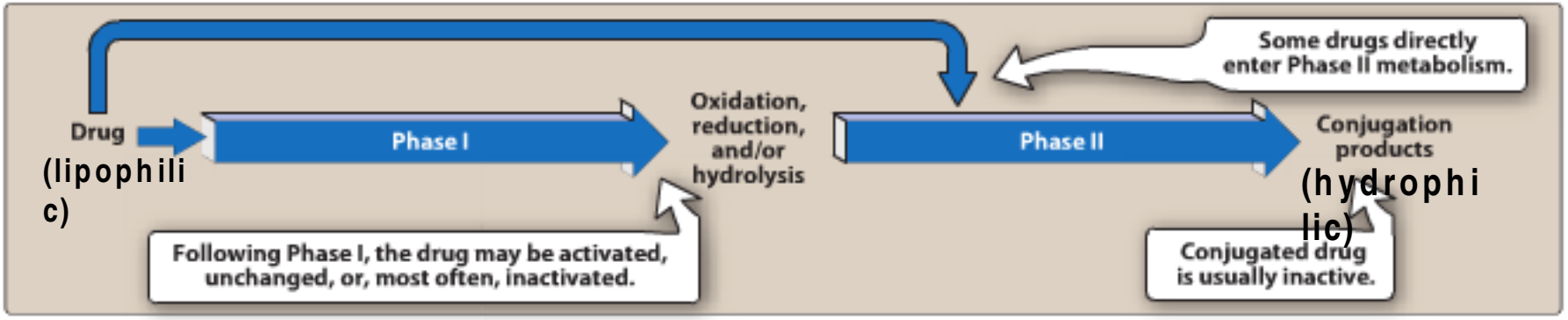


Figure 1.17
The biotransformation of drugs.

1. Phase I:

- Phase I reactions convert lipophilic molecules into more polar molecules by:

  introducing or **unmasking a polar functional group** such as –OH or –NH2.

- Pharmacologic activity after phase I metabolism may:
  - Increase
  - Decrease
  - Unchanged

Phase I metabolism

-Metabolizes drugs to creates sites for phase II metabolism

1) Oxidation (adds O) typically via simple addition of O or hydroxylation (adds H and O). Mediated predominantly via microsomal endoplasmic reticulum cytochrome P450 liver enzymes. Kidney and nervous tissue enzymes can also oxidize compounds

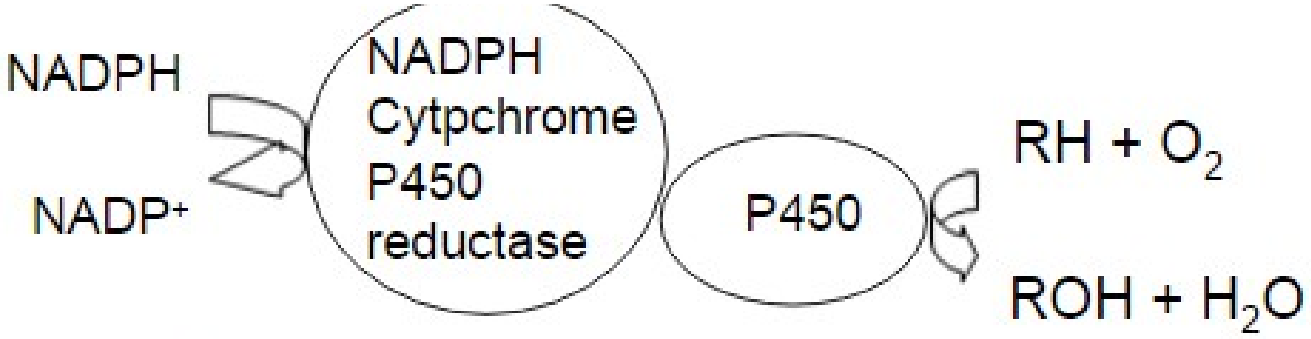
2) Reduction (gain of electron or H). Mediated by P450 enzymes

3)Hydrolysis (addition of water). Performed by hydrolytic enzymes called plasma esterases e.g. plasma cholinesterase

a.Phase I reactions utilizing the P450 system:

- Are catalyzed by the cytochrome P450 system (microsomal mixed-function oxidases)
- The most common reaction catalyzed by cytochromes P450 is **a monooxygenase reaction**

  e.g., insertion of one atom of oxygen into an organic substrate (RH) while the other oxygen atom is reduced to water:



RH – drug substance

1) Nomenclature:

- The enzymes themselves, are designated with the abbreviation CYP
- Followed by a number indicating the family **(3)**
- A capital letter indicating the subfamily **(A)**
- Another numeral for the individual gene or specific enzymes **(4)**
- Example: CYP3A4.

2) Specificity:

- Many different P450 isoforms.
- They can modify a large number of structurally diverse substrates.
- One drug may be a substrate for more than one isozyme.
- Four isozymes are responsible for the vast majority of P450-catalyzed reactions.
  - CYP3A4/5
  - CYP2D6
  - CYP2C8/9
  - CYP1A2
- Considerable amounts of CYP3A4 are found in intestinal mucosa (responsible for first-pass metabolism)

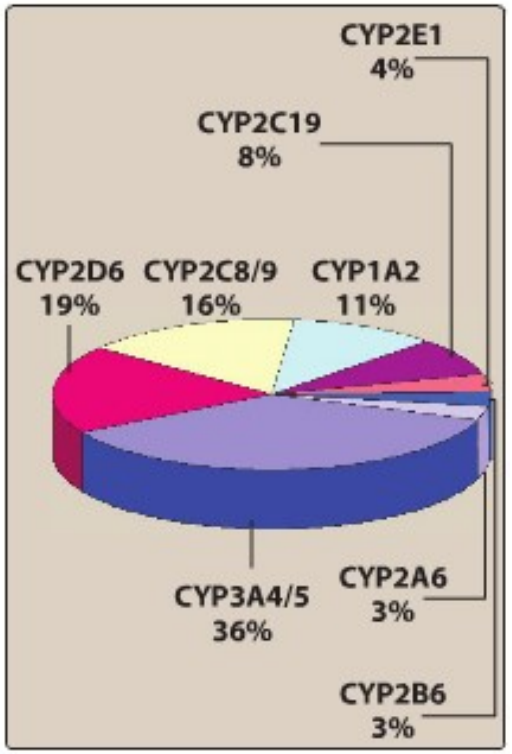
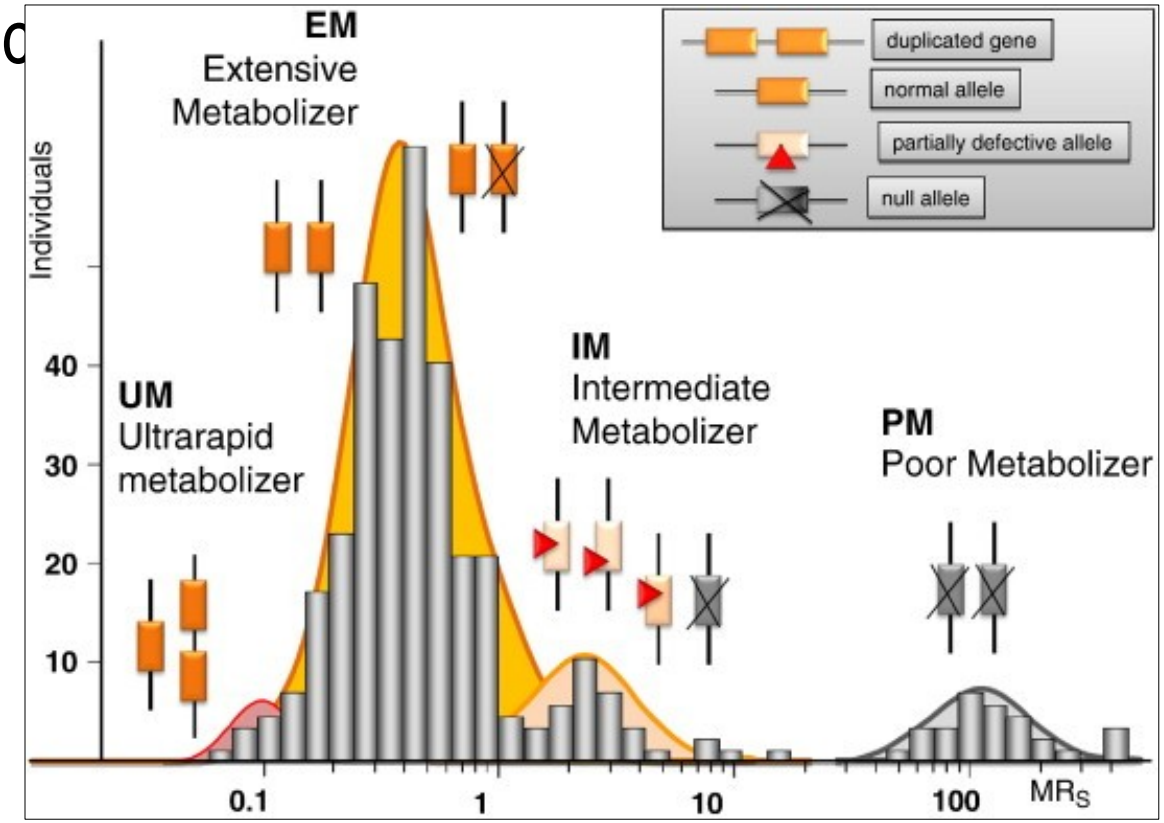


Figure 1.18
Relative contribution of cytochrome P450 (CYP) isoforms to drug bio-transformatin.

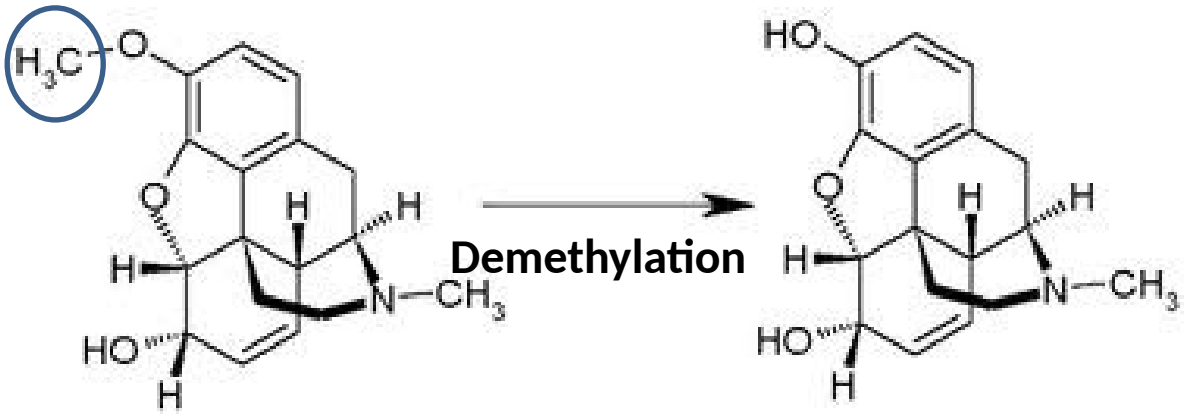
3) Genetic variability:

- P450 enzymes exhibit considerable genetic variability among individuals and racial groups.
- Variations in P450 activity may alter a drug's effi cac



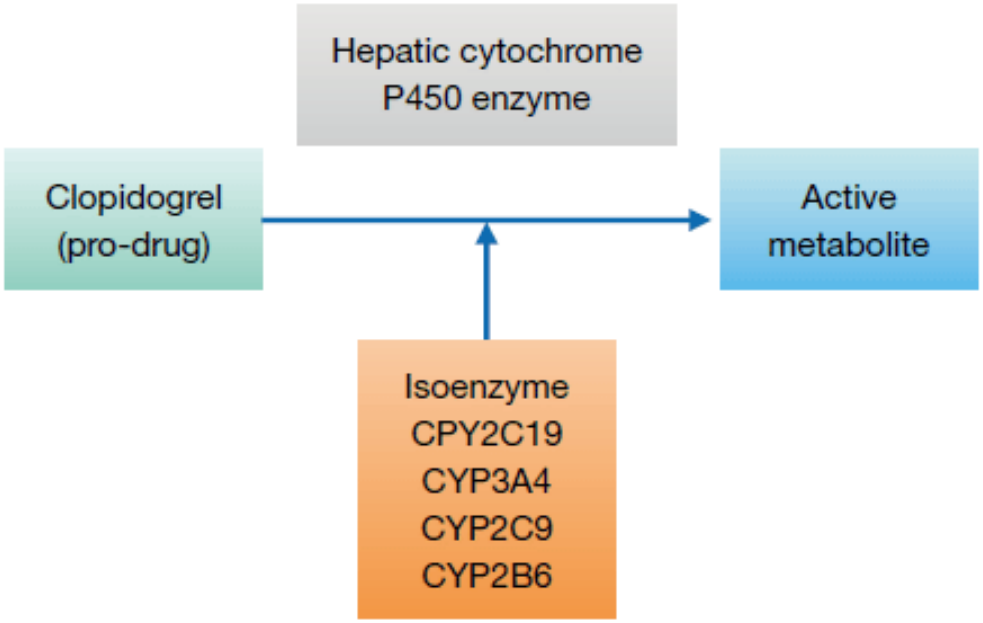
Example 1:

- **CYP2D6** exhibits genetic polymorphism.
- Patients lacking CYP2D6 can not benefit from **codeine** analgesic effect
- Codeine is activated by O-demethylation by CYP2D6 enzyme
- This polymorphism is in part **racially determined**:
  - 5 to 10 percent in European Caucasians
  - 2 percent of Southeast Asians



Example 2:

- **CYP2C** subfamily of isozymes.
- **Clopidogrel** is CYP2C19 substrate
- Poor CYP2C19 metabolizers have a higher incidence of cardiovascular events (for example, stroke or myocardial infarction) when taking clopidogrel.



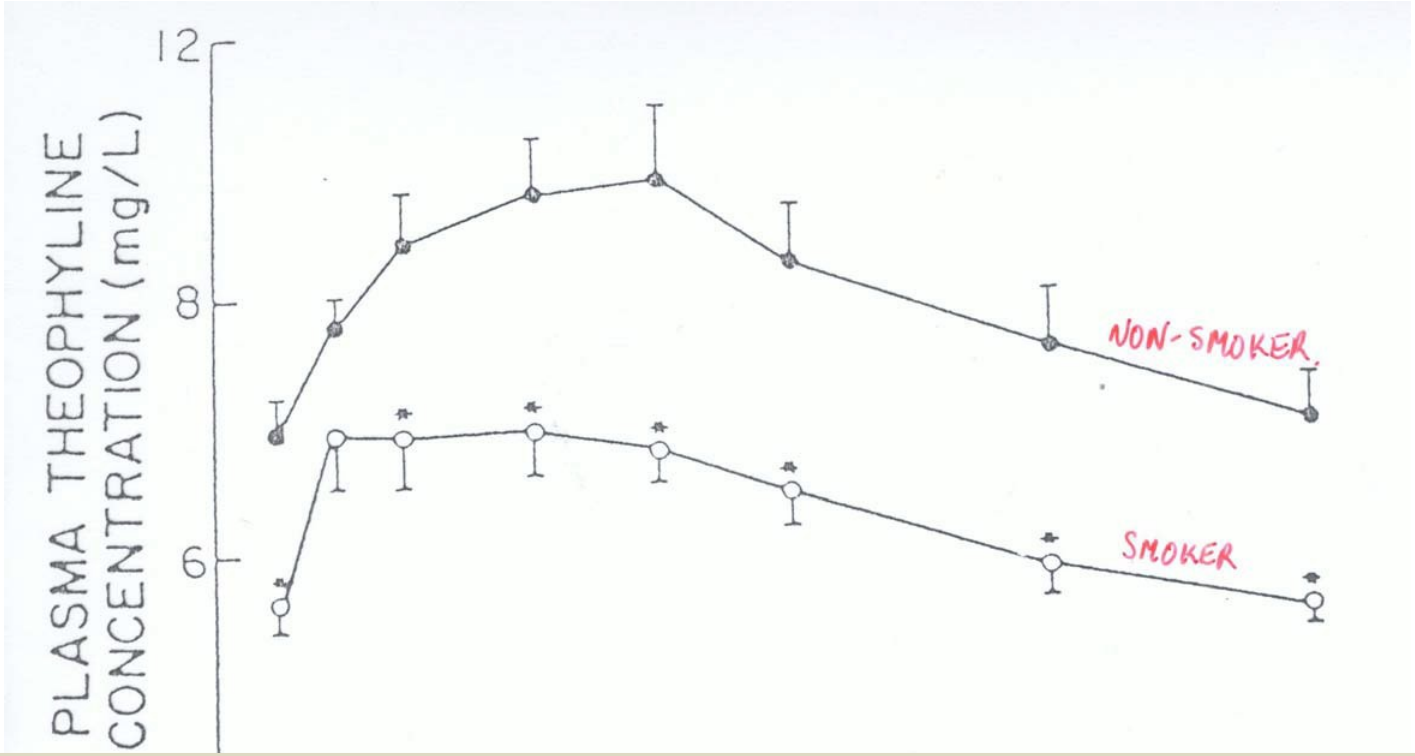
4) Inducers:

- Enzyme synthesis initiated within 24 h of exposure, increasing over 3 –5 days
- Effect decreases over 1 –3 weeks after inducing agent is discontinued
- **Environmental Factors and xenobiotics:**
  - Cigarette smoking, high protein diet, ethanol, exposure to insecticides (DDT, Lindane) & polychlorinated biphenyl (PCBs)
- **Certain drugs:**
  - Barbiturates, phenytoin, carbamazepine, rifampicin& dexamethasone

Consequence of increasing drug metabolism

1. Decrease plasma drug concentrations
2. Decrease drug activity if the metabolite is inactive
3. Increase drug activity if the metabolite is active
4. Decrease therapeutic drug effect

The effect of smoking on theophylline metabolism



If you know that the daily dose of theophylline is 13mg/kg, what will you expect to happen if the smoker patient took the same dose, explain?

5) Inhibitors:

- Inhibition of CYP isoenzyme activity is an important source of drug interactions
- Forms of inhibition:
  - Through competition for the same isoenzyme.
  - Inhibiting reactions for which they are not substrates (e.g., ketoconazole)

Important CYP inhibitors are:

- **Drugs:** Erythromycin, Ketoconazole, Ritonavir, Cimetidine

- **Natural substances** may also inhibit drug metabolism.
  - **grapefruit** and its juice **inhibits CYP3A4**
    - Examples on drugs that are substrates for this enzyme are:
      - Nifedipine (CCBs)
      - Clarithromycin (Macrolide antibiotic)
      - Simvastatin (Lipid lowering agent)

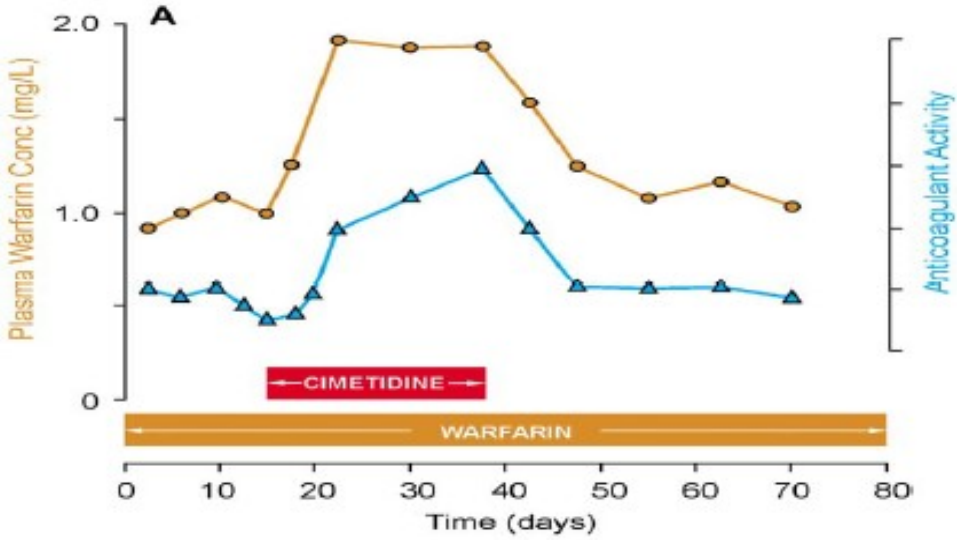
Example:

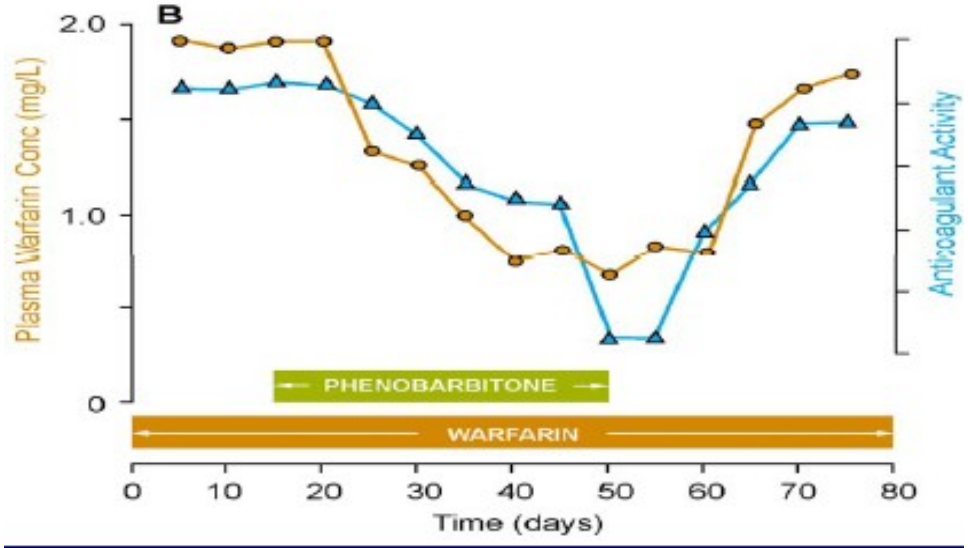
Warfarin biotransformation may be inhibited by omeprazole

- **Omeprazole** is a potent inhibitor of three of the CYP isozymes responsible for **warfarin metabolism**

- If the two drugs are taken together, plasma concentrations of **warfarin increase**, which leads to greater inhibition of coagulation and risk of hemorrhage and other serious bleeding reactions.

Inhibition/induction of warfarin metabolism, examples ...





- The effect of co-administration of the **inhibitor cimetidine** (A), on steady-state plasma warfarin concentration and its anti-coagulant effect
- The **inducer phenobarbitone** (B) on steady-state plasma warfarin concentration and its anti-coagulant effect

b. Phase I reactions not involving the P450 system:

- These include:
  - amine oxidation
    - e.g., oxidation of catecholamines or histamine
  - alcohol dehydrogenation
    - e.g. ethanol oxidation
  - esterases
    - e.g., metabolism of pravastatin in liver
  - hydrolysis
    - e.g., of procaine

2. Phase II (conjugation of endogenous molecule with drug)

1. **If the metabolite from Phase I metabolism is sufficiently polar, it can be excreted by the kidneys.**
2. **many Phase I metabolites are too lipophilic**



conjugation with an endogenous substrate

glucuronic acid, sulfuric acid, acetic acid and an amino acid



polar, more water-soluble compounds most often therapeutically inactive.

A notable exception is morphine-6-glucuronide , which is more potent than morphine.

Glucuronidation:

- Most common phase 2 reaction
- Most important conjugation reaction.
- Neonates are deficient in this conjugating system
  - Chloramphenicol

    is inactivated by the addition of glucuronic acid, resulting in gray baby syndrome.
  - Billirubin

- Drugs already possessing an –OH, –NH2, or –COOH group may enter Phase II directly and become conjugated without prior Phase I metabolism.

- The highly polar drug conjugates may then be excreted by the kidney or in bile.

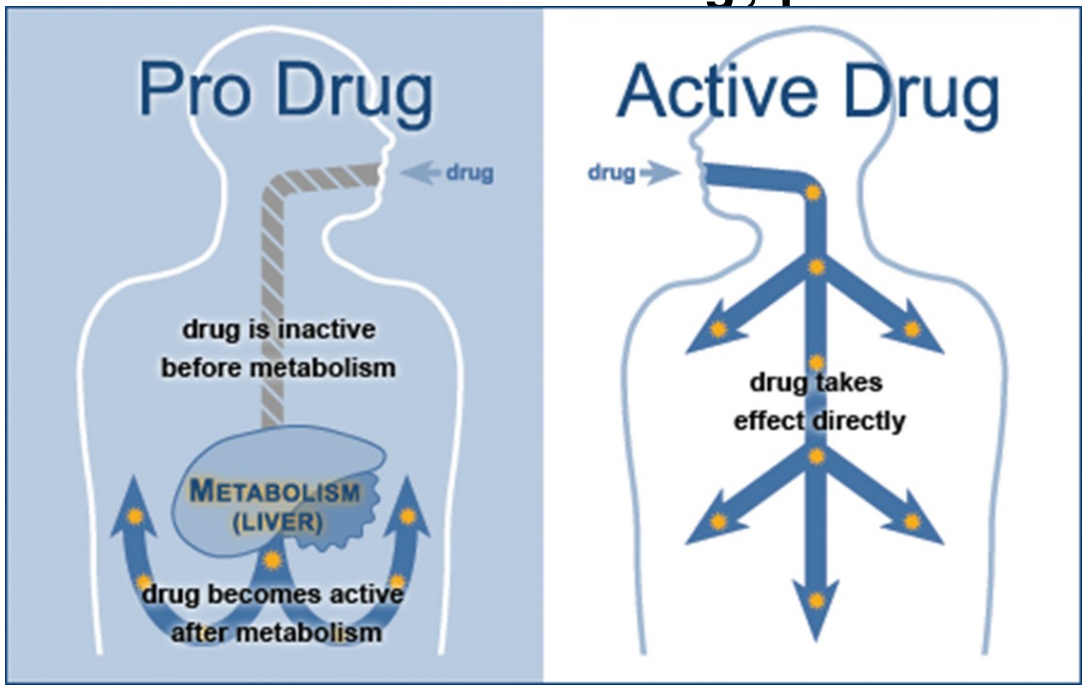
3. Reversal of order of the phases:

- Not all drugs undergo Phase I and II reactions in that order.

- For example, isoniazid is first acetylated (a Phase II reaction) and then hydrolyzed to isonicotinic acid (a Phase I reaction).

Outcomes of metabolism:

1. Abolishes activity and terminates drug action
2. Can promote activity -prodrug–**eg, acetylsalicylate**
3. No change in activity –**eg, diazepam ---> nordiazepam**
4. Produce toxic metabolites – **eg, paracetamol**



IV. DRUG EXECRETION

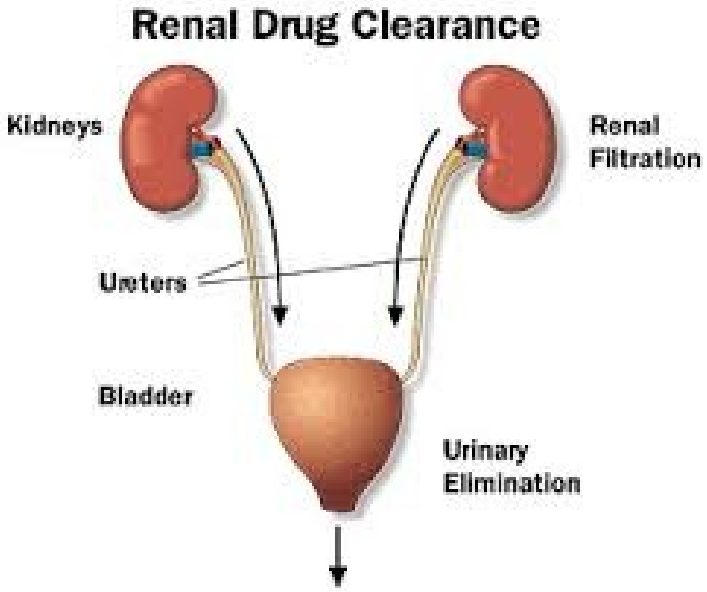
Routes of excretion

- **Main routes of excretion**
  1. Renal excretion
  2. Biliary excretion
- **Others routes of excretion**
  1. Exhaled air
  2. Salivary
  3. Sweat
  4. Milk
  5. Tears

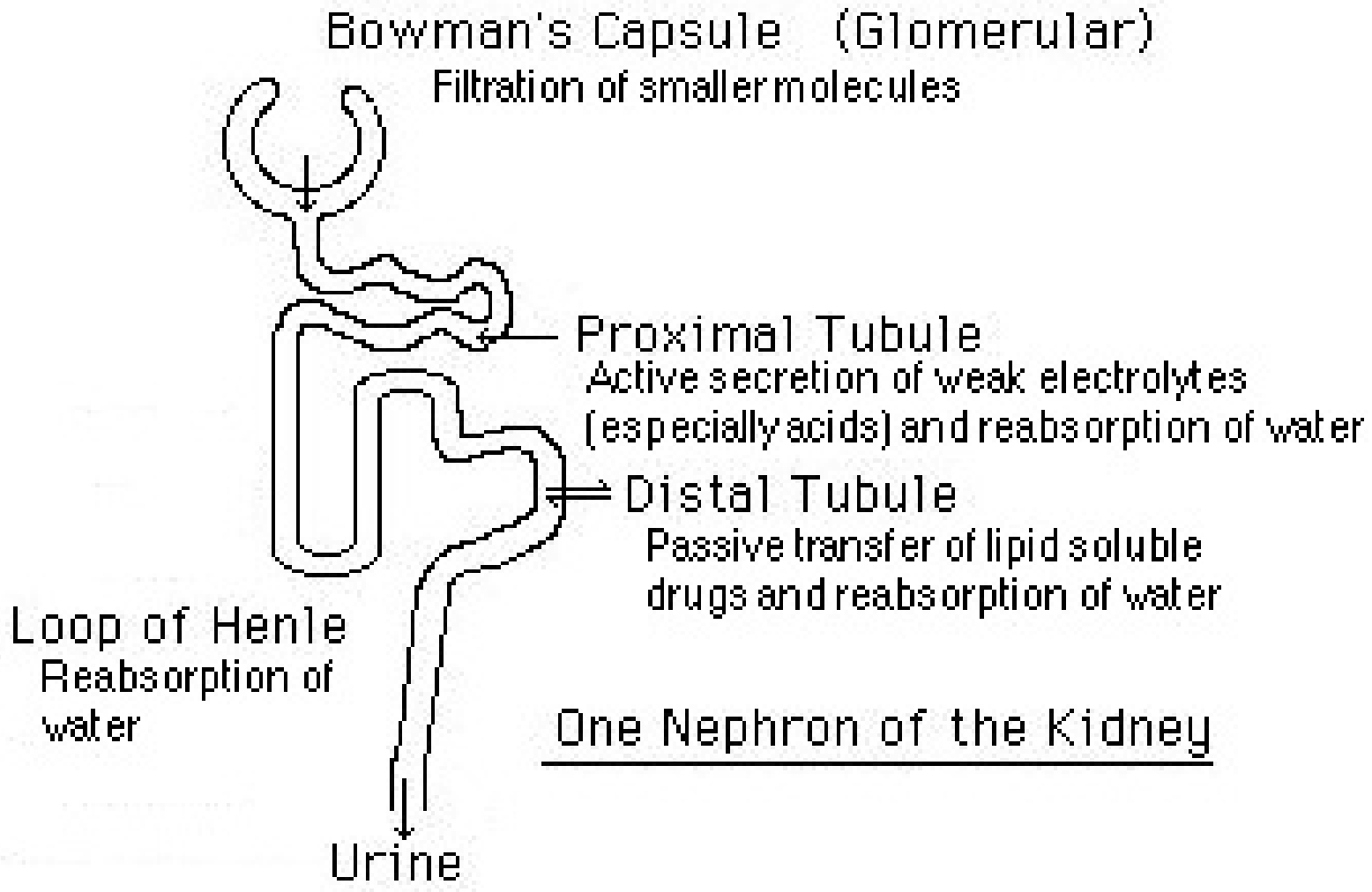
Renal excretion

Renal excretion

- The main organ for drug excretion is the **kidney**
- The functional unit of the kidney is the **nephron**
- There are three major processes:
  1. Glomerular filtration
  2. Proximal tubular secretion
  3. Distal tubular reabsorption



Nephron structure



A. Renal elimination of a drug

- Elimination of drugs via the kidneys into urine involves the three processes:

1. Glomerular filtration:

- The normal glomerular filtration rate is125 mL/min
- Passage of drugs into the glomerular filtrate influenced by:
  - Glomerular filtration rate
  - Protein binding of the drugs
  - No effect of lipophilicity and pH

2. Proximal tubular secretion:

- Secretion in the proximal tubules occurs through **active transport**:
  - One for anions (e.g., deprotonated weak acids, A^-)
  - One for cations (e.g., protonated weak bases, BH^+)
- They show low specificity
- Can transport many compounds
- Competition between drugs for these carriers can occur within each transport system.
- Example: Penicillins and probenecid

Note:

- Premature infants and neonates have an incompletely developed tubular secretory mechanism
- May retain certain drugs in the glomerular filtrate

3. Distal tubular reabsorption:

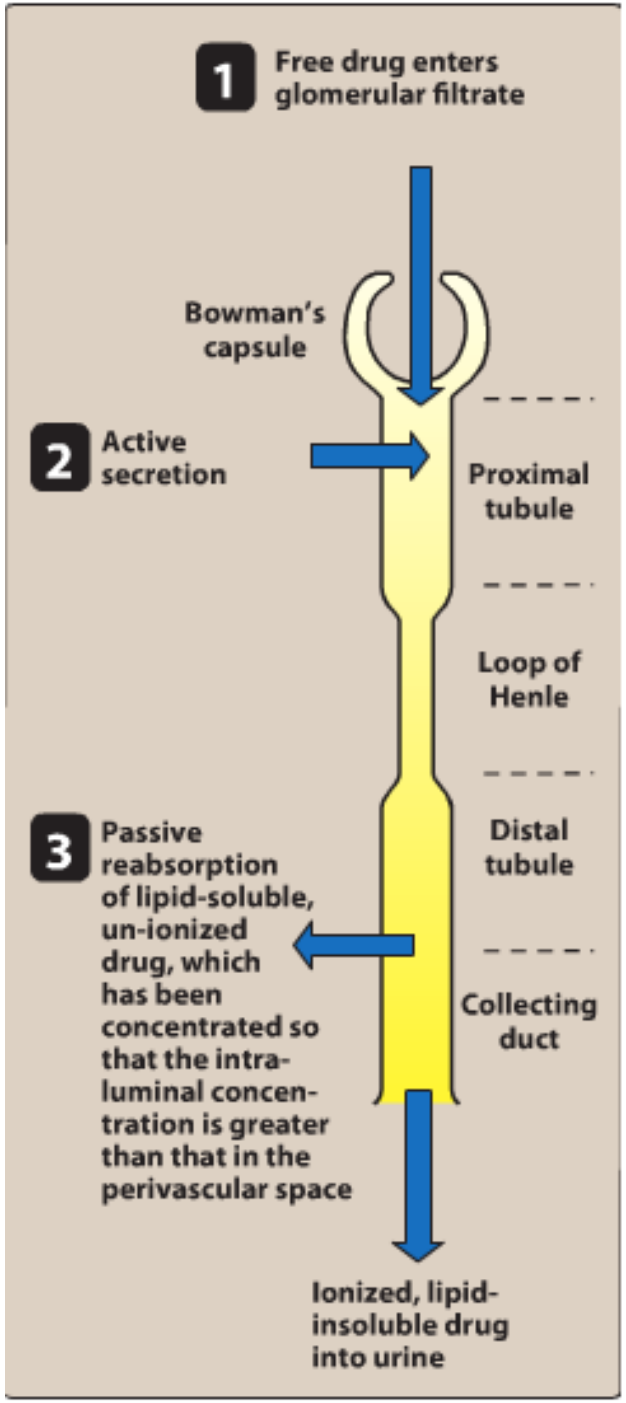
- Passive reabsorption of lipid soluble and uncharged drugs
- Urine pH affect reabsorption of drugs
  - Increasing the ionized form of the drug decrease its back-diffusion and increase the clearance of an undesirable drug.

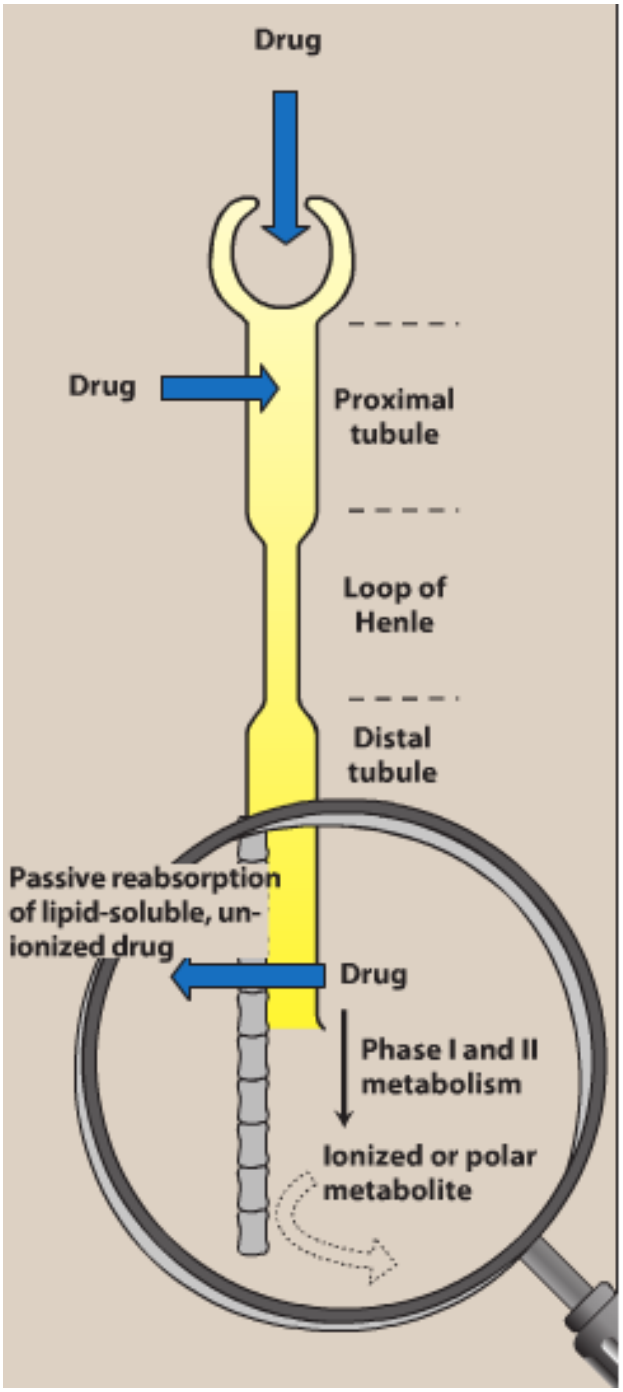
- Manipulating the pH of the urine "ion trapping."
  - Acidification of the urine increases elimination of weak bases
  - Alkalinization of the urine increases elimination of weak acids

Example:

a) An overdose of phenobarbital or acetylsalicylic acid (weak acids) can be treated with bicarbonate , which alkalinizes the urine and enhances the excretion of the ionized drug

b) An overdose of amphetamine (weak base) can be treated with NH_4Cl which acidifies the urine and increase the protonated form of drug (BH^+) and enhance its renal excretion


1
Free drug enters glomerular filtrate
Bowman's capsule
2
Active secretion
Proximal tubule
Loop of Henle
Distal tubule
3
Passive reabsorption of lipid-soluble, un-ionized drug, which has been concentrated so that the intra-luminal concen-tration is greater than that in the perivascular space
Collecting duct
Ionized, lipid-insoluble drug into urine


Drug
Drug
Proximal tubule
Loop of Henle
Distal tubule
Passive reabsorption of lipid-soluble, un-ionized drug
Drug
Phase I and II metabolism
Ionized or polar metabolite

CLEARANCE BY OTHER ROUTES

- Other important routes of drug clearance
  - Intestines
  - Bile
  - Lungs
  - milk in nursing mothers
- The feces are primarily involved in elimination of:
  - Unabsorbed orally ingested drugs
  - Drugs that are secreted directly into the intestines or in bile.
- The lungs are primarily involved in the elimination of:

Total body clearance

- **The total body clearance** (CL_{total}): is the sum of the clearances from the various drug-metabolizing and drug-eliminating organs
- $\mathbf{CL_{total}} = \mathbf{CL_{hepatic}} + \mathbf{CL_{renal}} + CL_{pulmonary} + Cl_{other}$

- CL_{total} and $t_{1/2}$ are important measures of drug clearance that are used to:
  - Optimize drug therapy
  - Minimize toxicity

Factors altering renal drug clearance

Renal drug clearance is lower therefore you must reduce the dose in:

1. Elderly and newborn
2. Women (-20%) than men
3. Kidney and heart disease
4. Patients taking drugs which block secretion (aspirin, probenecid, cimetidine)

Clinical situations resulting in changes in drug half-life

- **Dose Adjustment** is required
  - When a patient has an abnormality that alters the half-life of a drug

- It is important to be able to predict in which patients a drug is likely to have a change in half-life.

The half-life of a drug is increased by :

1) Diminished renal or hepatic blood flow as: cardiogenic shock, heart failure, or hemorrhage
2) Decreased ability to extract drug from plasma as seen in renal disease
3) Decreased metabolism
   - Another drug inhibits the biotransformation
   - In hepatic insufficiency, as cirrhosis



Lower doses or less frequent dosing interval may require

The half-life of a drug may decrease by:

1) Increased hepatic blood flow
2) Decreased protein binding
3) Increased metabolism



Higher doses or more frequent dosing interval may require

Dosage Regimens

- **A dosage regimen:**

  is defined as the manner in which a drug is taken

- **An optimal dosage regimen is to:**
  - **Maximize benefit of the drug**
  - **Minimize adverse effects**

- To initiate drug therapy a dosage regimen is administrated either by **continuous infusion** or in **intervals of time and dose**
- The regimen depends on various patients and drug factors, including the rapidly a **steady state** must be achieved
  - Steady state: the state at which the rate of administration equals the rate of elimination
- The regimen is refined to achieve the optimal dosage regimen

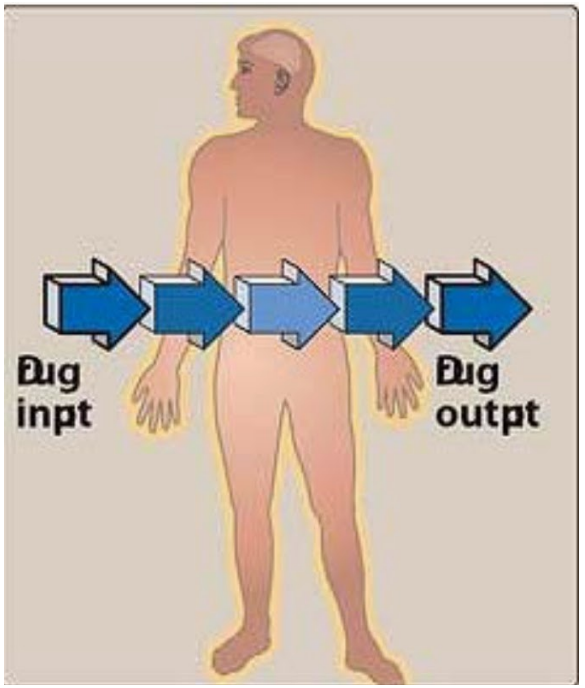
- **Drug administration**
  - Single dose
  - Continuous administration
    - IV infusion
    - Fixed-dose/fixed-time interval regimens

Dosage regimens

- IV infusion or oral fixed-dose/fixed-time interval regimens
- The drug accumulates until a steady state occurs
- At steady state the amount of drug administered equals the amount being eliminated
- At steady state:
  - The plasma and tissue levels remain constant with IV infusion and fluctuate around a mean in oral fixed dosage

Plasma concentration of a drug following IV infusion

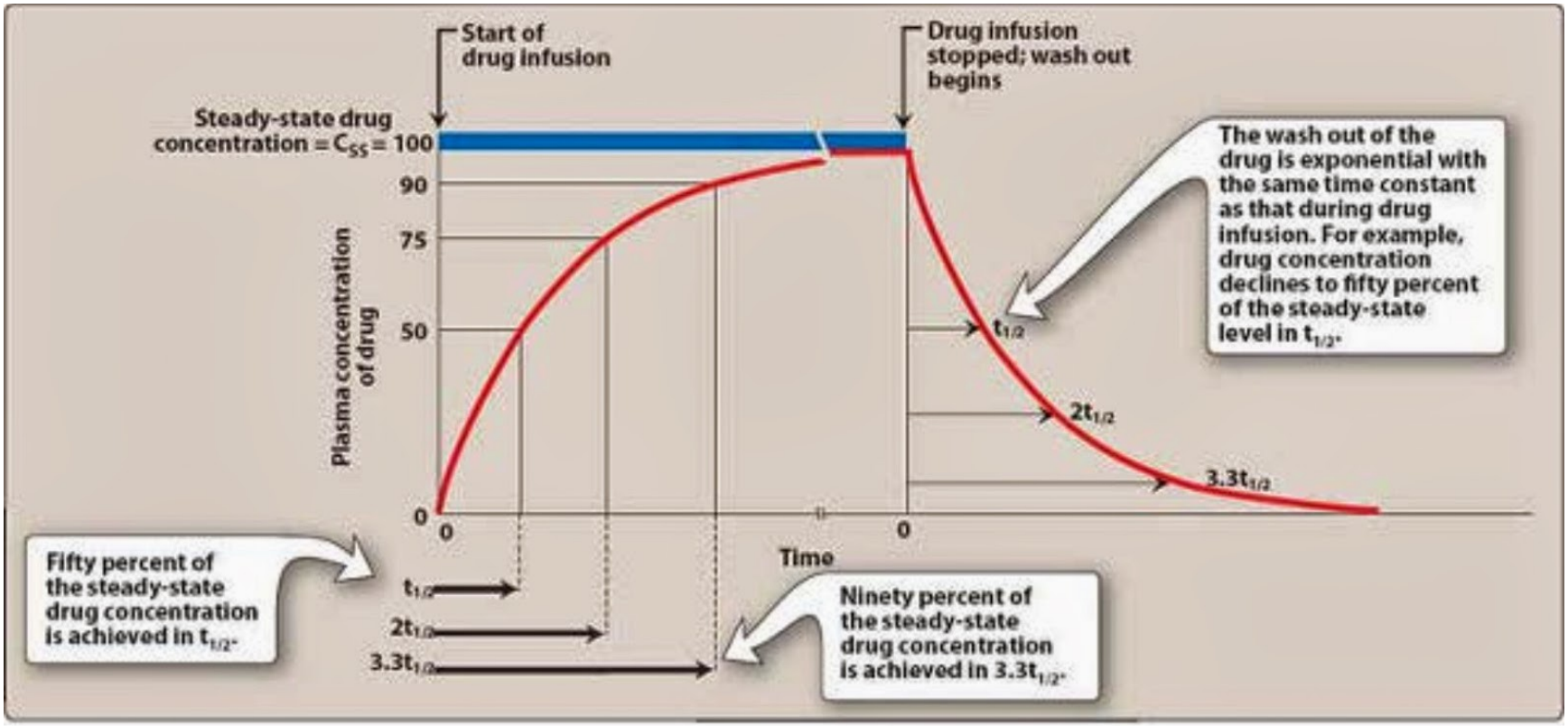
- Following initiation of IV infusion, the plasma concentration of a drug rises until the rate of drug eliminated from the body balances the input rate



steady state is achieved

- The plasma concentration of the drug remains constant (1st

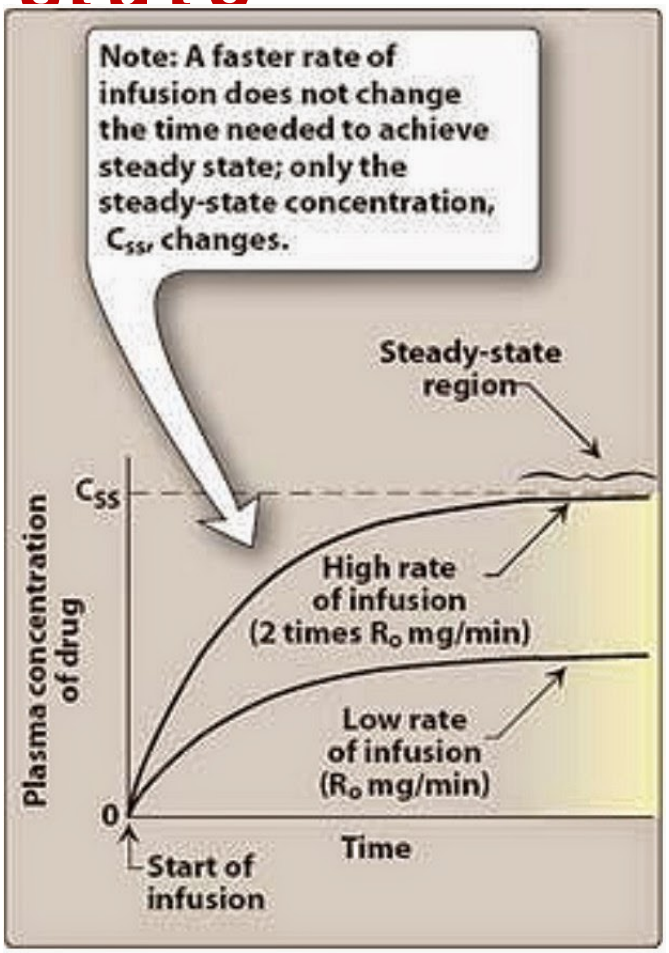
Time required to reach C_{ss}



Influence of the rate of drug infusion on the steady state

- The steady state plasma concentration (C_{ss}) is directly proportional to the infusion rate

- The C_{ss} is inversely proportional to the clearance of the drug
  - Hepatic or renal diseases can increase C_{ss}
  - Increased metabolism decreases C_{ss}



Effect of infusion rate on the steady-state concentration of drug in the plasma. (Ro = rate of infusion of a drug.)

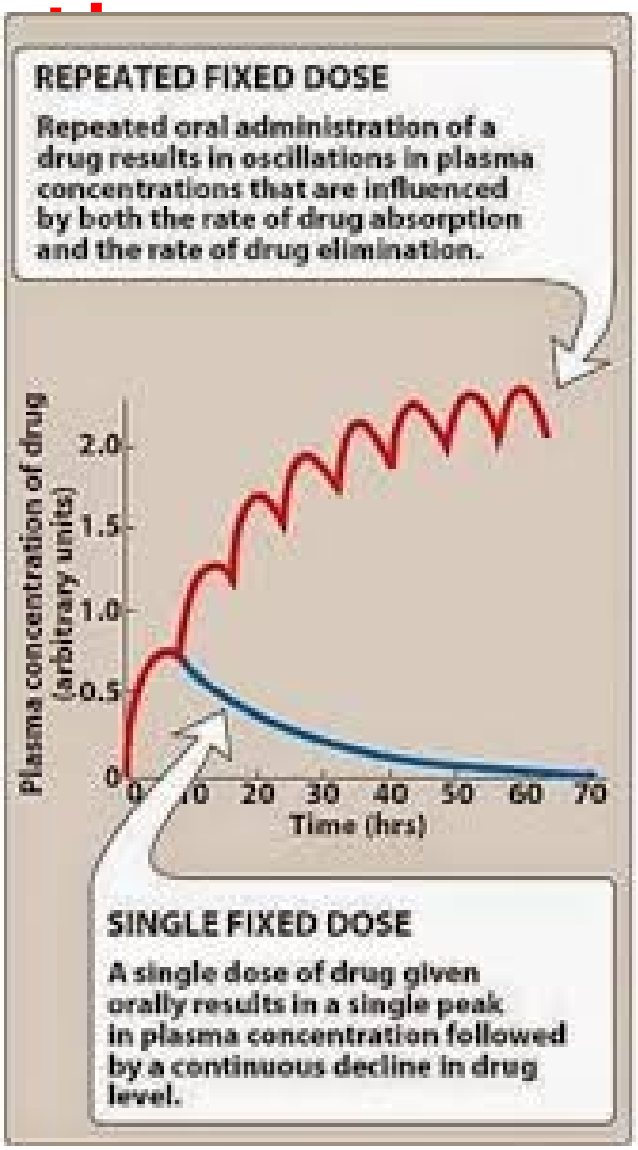
Fixed-dose/fixed-time-interval regimens

- Administration of a drug by fixed doses is more convenient than continuous infusion
- Fixed doses administered fixed-time intervals result in fluctuations in drug levels
- Fixed dose regimens
  - Multiple IV injections
  - Multiple oral administrations

Multiple oral administra

- Orally administered drugs may be absorbed slowly

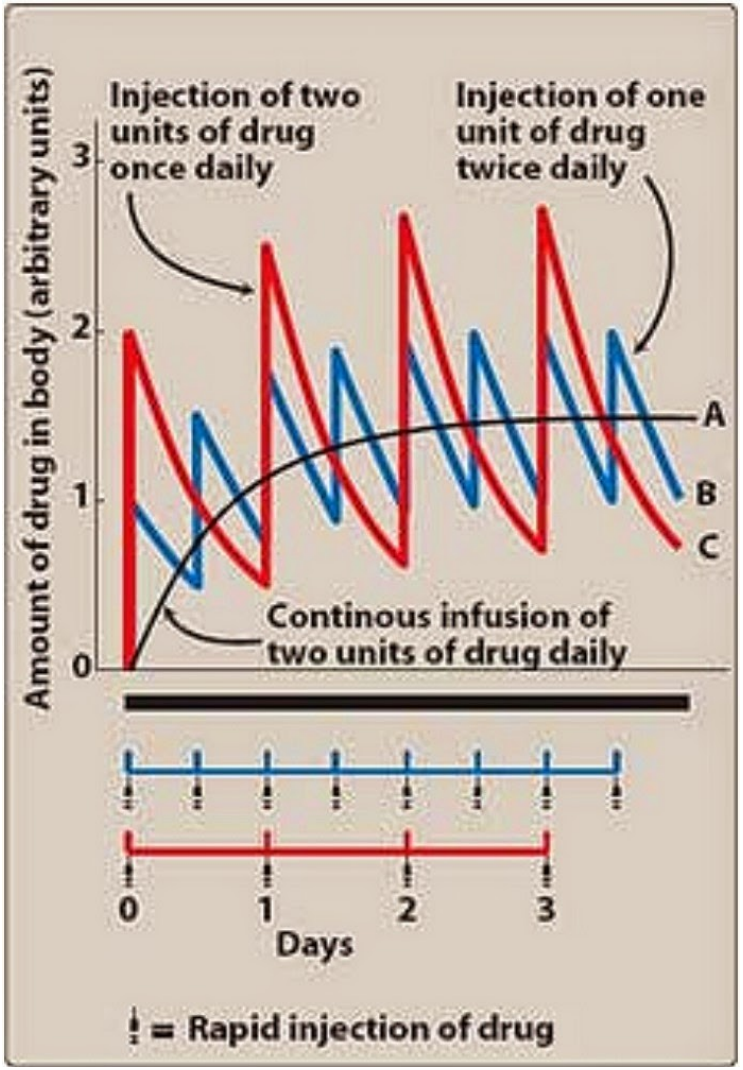
- Plasma concentration of the drug is influenced by
  - Rate of absorption
  - Rate of drug elimination



Multiple IV injections

- When a drug is given repeatedly at regular intervals, the plasma concentration increases until a steady state is reached

- The Using smaller doses at shorter intervals dose not change the at which the steady-state is approached



Optimization of dose

- The goal of drug therapy is to:
  - Achieve an maintain concentration within therapeutic window
  - Minimize toxicity and/or side effects
- Drugs of small therapeutic window (e.g., digoxin, warfarin):
  - Caution should be taken in selecting dosage regimen
  - Monitoring of drug levels may be needed
- Drug regimens are administered as:
  - Maintenance dose with/without loading dose

Maintenance dose/loading dose

- To maintain the plasma concentration within a specified range over long periods of therapy (C_{ss}) **maintenance doses** is used.
- If it is necessary to achieve the target plasma level rapidly, a **loading dose** is used
- **Loading dose**: is a higher single dose or a series of doses given to achieve the desired plasma level rapidly
- Loading dose may be associated with risk of drug toxicity

- Ideally, the dosing plan is based on knowledge of:
  - Minimum therapeutic level
  - Minimum toxic concentrations for the drug
  - Clearance
  - Vd
- **Dosing rate** = target C plasma* Cl/F
- **Loading dose** = (Vd)x (desired C_{ss})/F
- Loading dose is usefull for drugs relatively having long $t_{1/2}$

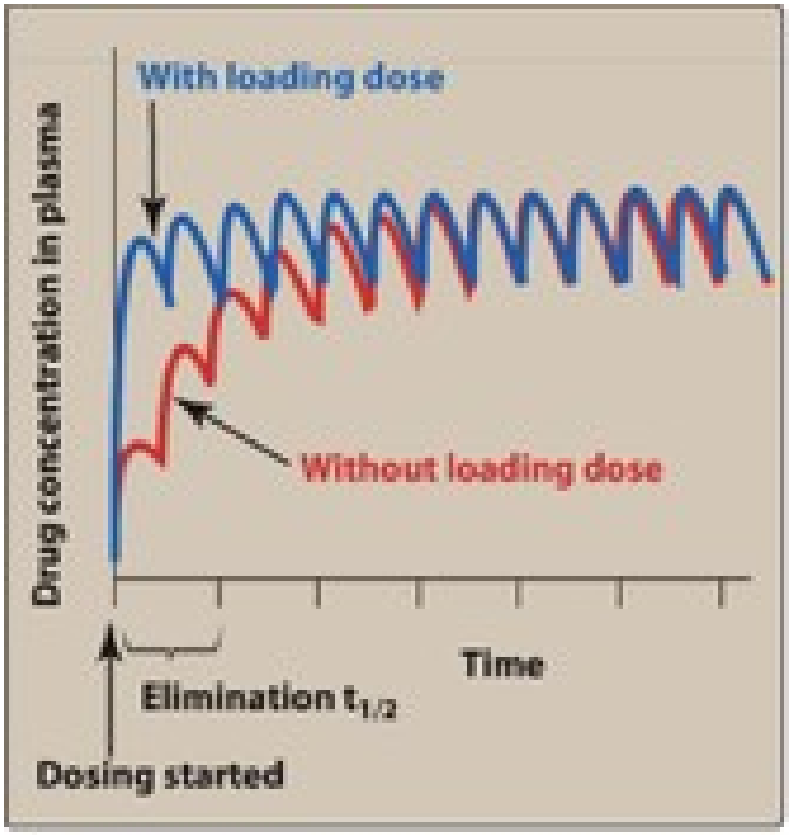


Figure 1.27
Accumulation of drug administered orally without a loading dose, and with a single oral loading dose administered at t=0.

Therapeutic Window

- The therapeutic window is the safe range between the minimum therapeutic concentration and the minimum toxic concentration of a drug.

- The concept is used to determine the acceptable range of plasma levels when designing a dosing regimen.

Therapeutic Window

